

SingleRAN

NSA-SA Selection Based on User Experience Feature Parameter Description

Issue

Draft B

Date

2026-01-30



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1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Renamed "SA-to-NSA switching based on downlink user experience" as "SA-to-NSA switching based on downlink traffic volume". For details, see [3 Overview](#) and [4 NSA/SA Selection Based on Downlink User Experience](#).
Revised the descriptions of the principles for NSA-to-SA switching based on downlink user experience. For details, see [4.2.1.1 Cell Edge Scenario](#).
Revised the license requirements for NSA/SA selection based on downlink user experience. For details, see [4.1.3.1 Licenses](#) and [4.2.3.1 Licenses](#).
Added the description of UEs for which real-time uplink coverage detection is not performed and revised the description of the procedure for the gNodeB to identify UEs under weak uplink coverage. For details, see [5.1.1.1 Identifying UEs Under Weak Uplink Coverage](#).
Revised the description of the principle for determining a UE in the low traffic volume state. For details, see [4.1.1.1 gNodeB Identifying High-Traffic UEs](#).
Revised descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 04 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for NSA-to-SA switching based on downlink user experience. For details, see: • 3 Overview • 4 NSA/SA Selection Based on Downlink User Experience • 4.2 NSA-to-SA Switching Based on Downlink User Experience	Modified parameters on the NR side: • Added the DL_NSA_TO_SA_SWITCH_SW option to the gNodeBParam.NetworkingOptionOptSw parameter. • Added the NR_TO_LTE_BIG_PKT_STA_TRANS_SW option to the gNodeBParam.NetworkingOptionOptSw parameter. Added parameters on the LTE side: • NrNFreq.SaPccFreqPriority • NsaDcMgmtConfig.DlNsaToSaSwitchA2RsrpThld • NsaDcMgmtConfig.NrCellIdleExperienceCoeff Modified parameters on the LTE side: • Added the DL_NSA_TO_SA_SWITCH_SW option to the EnodebAlgo	NR FDD Low-frequency NR TDD LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
	ExtSwitch.NsaDcAlgoExtSwitch parameter. Added the LARGE_PKT_UE_IDENT_SW option to the EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch parameter.		
Added cell-level control on the "NSA/SA selection based on downlink traffic volume" function. For details, see: 4.1.1 Principles 4.1.2.2 Impacts 4.1.4.1.1 Data Preparation 4.1.4.1.2 Using MML Commands	Modified parameter: Added the NSA_SA_DL_OPT_FRBD_SW option to the NRInterRatHoParam.MobilityHOControlSwitch parameter.	NR FDD Low-frequency NR TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added ANR for missing neighboring LTE cells. For details, see: • 4.1.1.2 SA-to-NSA Switching Based on UE Measurements • 4.1.2.2 Impacts • 4.1.3.2.1 Prerequisite Functions • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands • 5.1.1.2 SA-to-NSA Switching Based on UE Measurements • 5.2.2 Impacts • 5.3.2.1 Prerequisite Functions • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the NSA_SA_SEL_MISS_CELL_ANR_SW option to the NRCellAnr.AlgoSwitch parameter.	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added cell-level control on the "NSA/SA selection based on weak uplink coverage" function. For details, see: • 5.1 Principles • 5.2.2 Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the NSA_SA_UL_WEAK_FRBD_SW option to the NRInterRatHoParam.MobilityHOControlSwitch parameter.	Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the following counters: • N.DLHighTraffic.U E.Count: used to observe the number of times UEs with high traffic volume are identified in a cell. For details, see 4.1.4.2 Activation Verification. • N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.PrepAttOut.D L, N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.ExecAttOut.D L, and N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.ExecSuccOut. DL: used to observe whether NSA/SA selection based on downlink traffic volume has taken effect. For details, see 4.1.4.2 Activation Verification. • N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.PrepAttOut.U L, N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.ExecAttOut.U L, and N.HO.InterRAT.N2E.ResourceOpt.SA2 NSA.ExecSuccOut. UL: used to observe whether SA-to-NSA switching based on weak uplink coverage has taken effect. For details,	None	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
see 5.4.2 Activation Verification .			
Added incompatibility of BTS5929R with NSA/SA selection based on downlink user experience and NSA/SA selection based on uplink coverage. For details, see: • 4.1.3.3 Hardware • 5.3.3 Hardware	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
NR	MRFD-17126 2	NSA/SA Selection Based on User Experience (NR)	4 NSA/SA Selection Based on Downlink User Experience 5 NSA/SA Selection Based on Uplink Coverage
LTE FDD	MRFD-17122 2	NSA/SA Selection Based on User Experience (LTE FDD)	
LTE TDD	MRFD-17123 2	NSA/SA Selection Based on User Experience (LTE TDD)	

2.4 Differences

Differences Between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
NSA/SA selection based on downlink traffic volume	None	4 NSA/SA Selection Based on Downlink User Experience
NSA/SA selection based on uplink coverage	Supported only in LTE FDD	5 NSA/SA Selection Based on Uplink Coverage

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
NSA/SA selection based on downlink traffic volume	None	4 NSA/SA Selection Based on Downlink User Experience
NSA/SA selection based on uplink coverage	Supported only in NR TDD	5 NSA/SA Selection Based on Uplink Coverage

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
NSA/SA selection based on downlink traffic volume	Supported in NSA and SA hybrid networking, but not in SA networking or NSA networking	4 NSA/SA Selection Based on Downlink User Experience
NSA/SA selection based on uplink coverage	Supported in NSA and SA hybrid networking, but not in SA networking or NSA networking	5 NSA/SA Selection Based on Uplink Coverage

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
NSA/SA selection based on downlink traffic volume	Supported only in low frequency bands	4 NSA/SA Selection Based on Downlink User Experience
NSA/SA selection based on uplink coverage	Supported only in low frequency bands	5 NSA/SA Selection Based on Uplink Coverage

3 Overview

In NSA and SA hybrid networking of NR cells, UEs in connected mode that support both NSA and SA can access NR cells in NSA or SA mode. The networking mode affects user-perceived rates.

NSA or SA, whichever can provide better service experience for UEs, is selected based on the downlink user experience or uplink coverage of the UEs. The following functions are included:

- NSA/SA selection based on downlink user experience. For details, see [4 NSA/SA Selection Based on Downlink User Experience](#). This function consists of the following subfunctions:
 - SA-to-NSA switching based on downlink traffic volume, as shown in [Figure 3-1](#). For details, see [4.1 SA-to-NSA Switching Based on Downlink Traffic Volume](#).
 - NSA-to-SA switching based on downlink user experience, as shown in [Figure 3-2](#). For details, see [4.2 NSA-to-SA Switching Based on Downlink User Experience](#).
- NSA/SA selection based on uplink coverage. For details, see [5 NSA/SA Selection Based on Uplink Coverage](#). This function consists of the following subfunctions:
 - SA-to-NSA switching based on weak uplink coverage, as shown in [Figure 3-1](#). For details, see [5.1.1 SA-to-NSA Switching Based on Weak Uplink Coverage](#).
 - NSA-to-SA switching based on strong uplink coverage, as shown in [Figure 3-2](#). For details, see [5.1.2 NSA-to-SA Switching Based on Strong Uplink Coverage](#).

Figure 3-1 Diagram of SA-to-NSA switching based on downlink traffic volume or NSA/SA selection based on weak uplink coverage in NSA and SA hybrid networking

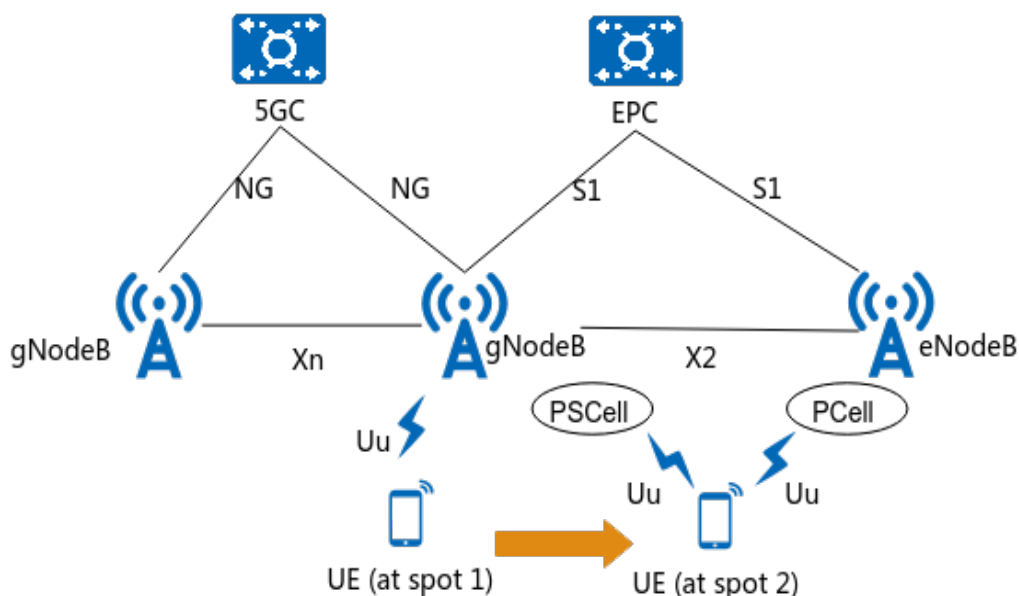
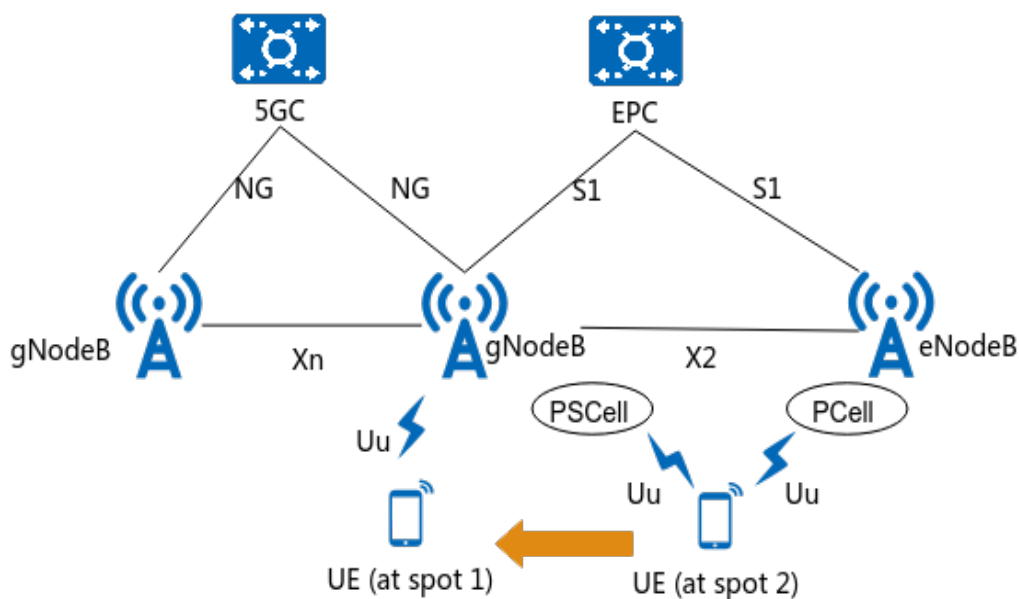


Figure 3-2 Diagram of NSA-to-SA switching based on downlink user experience or NSA/SA selection based on strong uplink coverage in NSA and SA hybrid networking



When both NSA/SA selection based on downlink user experience and NSA/SA selection based on uplink coverage are enabled, NSA/SA selection is evaluated preferentially based on uplink coverage. If the downlink traffic volume meets the trigger condition but the uplink coverage does not meet the trigger condition, NSA/SA selection is not performed, as described in [Table 3-1](#).

Table 3-1 Function effective status

Uplink Coverage Conditions Fulfilled		Uplink Coverage Conditions Unfulfilled	
Downlink user experience conditions fulfilled	Downlink user experience conditions unfulfilled	Downlink user experience conditions fulfilled	Downlink user experience conditions unfulfilled
NSA/SA selection based on uplink coverage preferentially takes effect.		NSA/SA selection does not take effect.	NSA/SA selection does not take effect.

4 NSA/SA Selection Based on Downlink User Experience

4.1 SA-to-NSA Switching Based on Downlink Traffic Volume

SA-to-NSA switching based on downlink traffic volume enables the gNodeB to identify high-traffic UEs and trigger SA-to-NSA switching for UEs based on user experience. This function offers a networking scheme that improves service experience for UEs, helping achieve optimal user experience.

This function includes two steps:

1. The gNodeB identifies high-traffic UEs. For details, see [4.1.1.1 gNodeB Identifying High-Traffic UEs](#).
2. The gNodeB initiates SA-to-NSA switching based on user experience for the high-traffic UEs. The following policies are currently supported, depending on the RSRP acquisition method:
 - **SA-to-NSA switching based on UE measurements:** The gNodeB decides and executes a handover based on the RSRP from UE measurement. For details, see [4.1.1.2 SA-to-NSA Switching Based on UE Measurements](#).
 - **SA-to-NSA switching with RSRP predicted based on virtual grids:** The RSRP predicted using the NL RSRP model is used instead of the RSRP from UE measurement for handover decision and execution. For details, see [4.1.1.3 SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids](#).

If both policies are enabled, SA-to-NSA switching with RSRP predicted based on virtual grids takes effect preferentially.

After a UE accesses an NSA network, **E-UTRAN to NG-RAN traffic steering or fast return from E-UTRAN to NG-RAN** allows the UE to quickly return to an SA network. For details, see [4.1.1.4 Handling After UE Access in NSA](#).

NSA/SA selection based on downlink traffic volume is enabled when all of the following conditions are met:

- On the gNodeB side:
 - The LTE-NR NSA DC function is enabled by setting the **NRCellAlgoSwitch.NsaDcSwitch** parameter to **ON**.
 - The **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch.InterRatServiceMobilitySw** parameter is selected to enable inter-RAT mobility from NG-RAN to E-UTRAN.
 - The **NSA_SA_DL_SEL_OPT_SW** options of the **gNBOperator.OperatorInterRatPolicySw** and **gNodeBParam.NetworkingOptionOptSw** parameters are selected to enable NSA/SA selection based on downlink user experience.
The **NSA_SA_DL_SEL_OPT_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter takes effect only when the **NSA_SA_DL_SEL_OPT_SW** option of the **gNBOperator.OperatorInterRatPolicySw** parameter is selected.
 - The **EUTRAN_HO_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter is selected.
 - The **NSA_SA_DL_OPT_FRBD_SW** option of the **NRInterRatHoParam.MobilityHOControlSwitch** parameter is deselected for the target cell. If this option is selected for the target cell, NSA/SA selection based on downlink traffic volume will not take effect.
 - (Optional) In the RAN sharing with common carrier scenario (where the **gNBSharingMode.gNBMultiOpSharingMode** parameter is set to **SHARED_FREQ**), inter-RAT mobility policies are configured based on the operator by selecting the **EUTRAN_HO_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter.
 - (Optional) In inter-gNodeB CA scenarios, NSA/SA selection based on user experience for inter-gNodeB CA UEs is enabled by selecting the **NSA_SA_SEL_INTER_GNB_CA_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter.
- On the eNodeB side:
The **LTE_FDD_NSA_SA_DL_SEL_OPT_SW** or **LTE_TDD_NSA_SA_DL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter is selected.

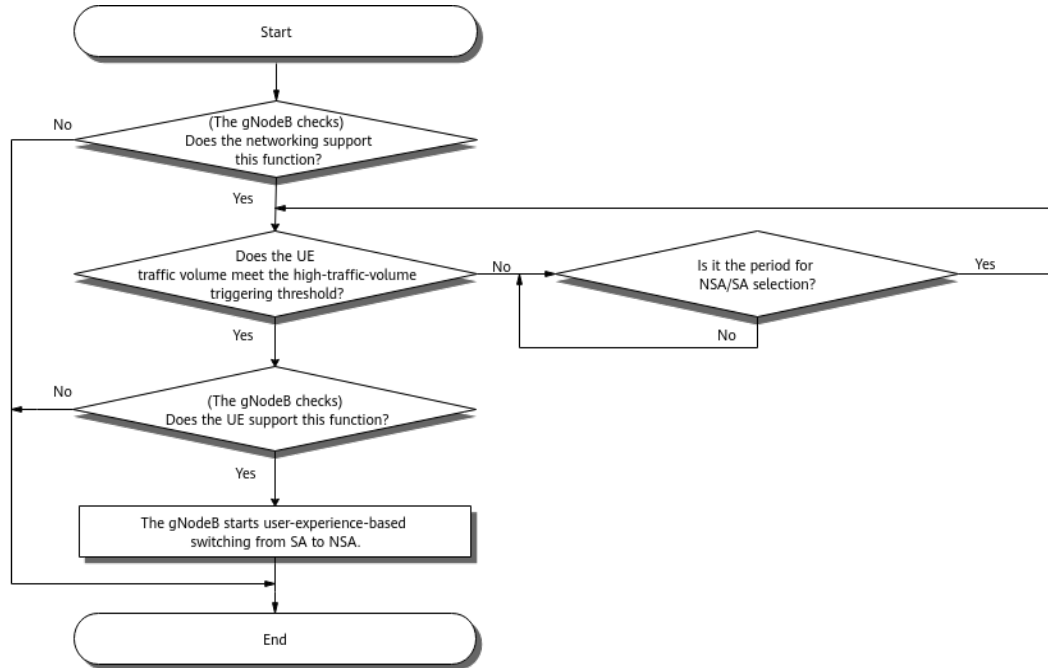
NSA/SA selection based on downlink traffic volume requires a UE to report its NR, EUTRA-NR, and EUTRA capabilities through a UECapabilityInformation message. If all these capabilities are included, the message length may exceed the maximum defined in 3GPP specifications. As a result, this message may be truncated, which can affect NSA/SA selection based on downlink traffic volume. When the UE capability separate query function is enabled (by selecting the **UE_CAP_SEPARATE_QUERY_SW** option of the **NRCellAlgoSwitch.ProcessSwitch** parameter), the gNodeB can query NR, EUTRA-NR, and EUTRA capabilities separately to ensure that the UE's complete capability information is obtained. This function supports blacklist control. When the **UE_CAPB_SEPARATE_QUERY_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlSwitch** parameter is selected, the preceding function does not take effect for blacklisted UEs.

4.1.1 Principles

4.1.1.1 gNodeB Identifying High-Traffic UEs

The procedure for the gNodeB to identify high-traffic UEs is shown in Figure 4-1.

Figure 4-1 Procedure for identifying high-traffic UEs



The procedure is as follows:

1. The gNodeB checks whether the networking supports this function
The NR cell must operate in NSA and SA hybrid networking; that is, the **gNBOperator.NrNetworkingOption** parameter is set to **SA_NSA**. Otherwise, none of the UEs in the NR cell supports this function, and the procedure ends.
2. The gNodeB checks whether the UE traffic volume meets the high-traffic-volume triggering threshold.
The gNodeB monitors the downlink traffic volume of all UEs in the cell in real time. The traffic volume of all UEs is initially low.
 - If the gNodeB detects that the downlink traffic volume for a UE is to be transmitted for a duration longer than or equal to the threshold specified by the **NRDUCellCarrMgmt.NsaSaSelOptHighTrafficThld** parameter, the UE enters the high traffic volume state. The gNodeB records the UE traffic volume as high and performs the next step.
 - If the gNodeB detects that the downlink traffic volume for a UE is to be transmitted for a duration shorter than or equal to the threshold specified by the **NRDUCellCarrMgmt.NsaSaSelOptLowTrafficThld** parameter, the UE enters the low traffic volume state, and the gNodeB records the UE traffic volume as low.
 - If none of the preceding conditions is met, the gNodeB keeps the original traffic volume state of the UE unchanged.

The gNodeB periodically selects high-traffic UEs in the cell at an interval specified by the **gNodeBParam.NsaSaSelOptPeriod** parameter.

3.

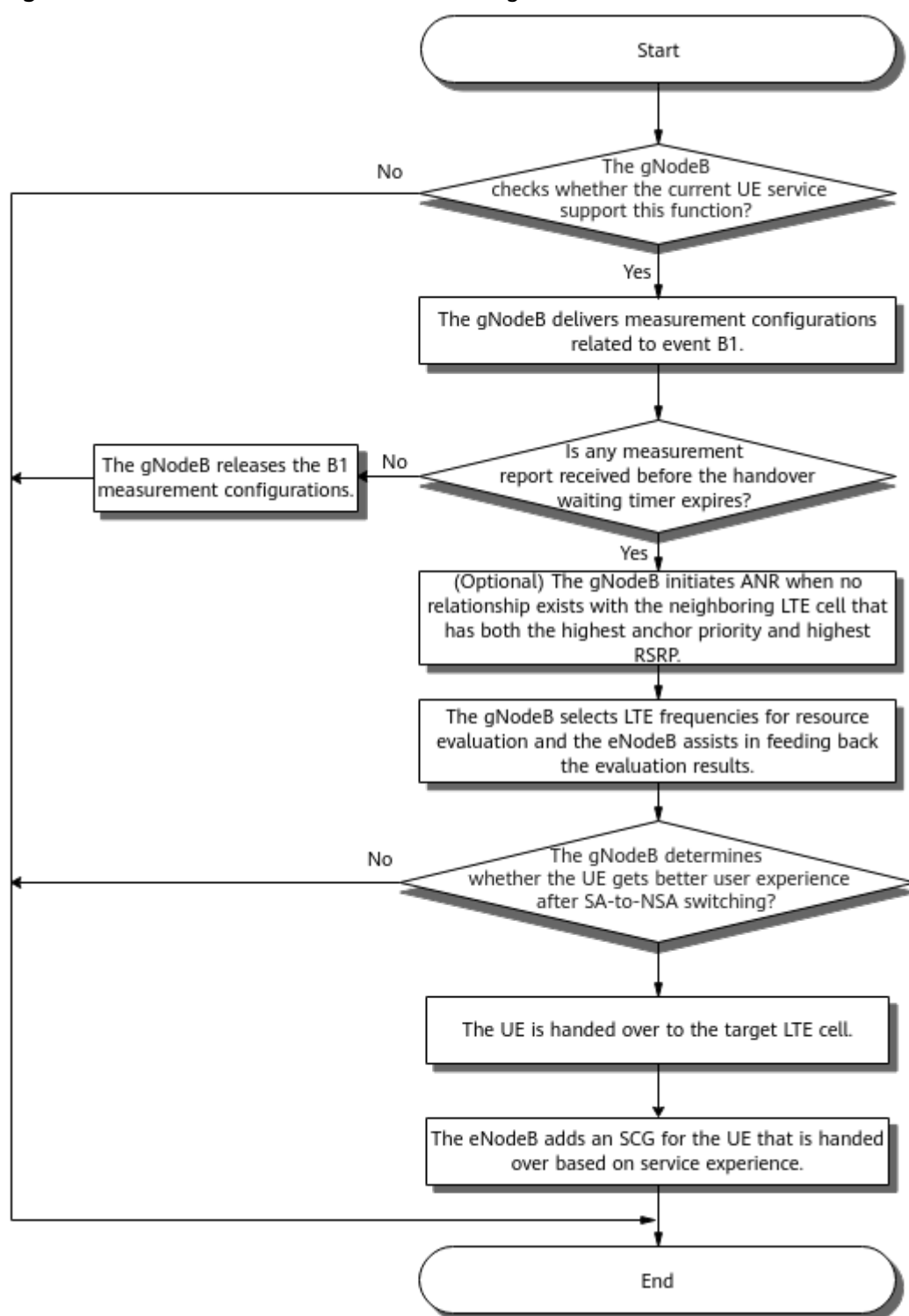
The gNodeB checks whether the UE complies with 3GPP Release 15 or later, as well as whether the UE supports SA, and handoverLTE-EPC and EN-DC capabilities. If yes, the procedure goes to the next step. If no, the procedure ends.
4.

The gNodeB starts the procedure for SA-to-NSA switching based on user experience for UEs in the high traffic state.

4.1.1.2 SA-to-NSA Switching Based on UE Measurements

The gNodeB starts SA-to-NSA switching based on UE measurements for UEs in the high traffic state, as shown in [Figure 4-2](#).

Figure 4-2 Procedure for SA-to-NSA switching based on UE measurements



The procedure is as follows:

1. The gNodeB checks the network option optimization policies for the QCI of services running on a UE.

The **gNBQciBearer.NetworkingOptionOptFlag** parameter specifies the policy. The following values are possible:

- **FORBID**: Network option optimization is not performed for a UE running services with the specified QCI.
- **PERMIT**: Network option optimization is allowed for a UE running services with the specified QCI. However, if services with a QCI for which this parameter is set to **FORBID** are also running on the UE, network option optimization is not performed for the UE.
- **FOLLOW**: Network option optimization is not performed for a UE running only the services with the specified QCIs. However, if services with other QCIs for which this parameter is set to a value other than **FOLLOW** are also running on the UE, network option optimization complies with the policies defined for such QCIs.

According to the UE based on the Networking Option Opt Flag settings corresponding to the QCIs of services running on the UE, the gNodeB takes the following actions:

- If the UE is running a service with a QCI for which the parameter is set to **PERMIT** and with no QCIs of services set to **FORBID**, the procedure goes to the next step.
- If the UE is running a service with a QCI for which the parameter is set to **FORBID** or running only services with QCIs for which the parameter is set to **FOLLOW**, the procedure ends, that is, NSA/SA selection based on downlink traffic volume does not take effect.

When flexible user steering is enabled (that is, when the **gNB RfSP Config.NetworkingOptionOptFlag** parameter is set to **TRUE** to allow customized network option optimization policies for UEs based on the RFSP), and RFSP-based NG-RAN-to-E-UTRAN handover takes effect (by selecting the **NR_TO_EUTRAN_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter), the customized network option optimization policies take effect.

2. The gNodeB delivers measurement configurations related to inter-RAT event B1.

The gNodeB includes the frequencies in the descending order of priority in the measurement configurations related to inter-RAT event B1 (namely that for network architecture selection), delivers them to the UE, and starts a handover waiting timer, whose length is always 2s. When the measured RSRP value of an E-UTRAN cell exceeds the *Thresh* (specified by the **NRCellHoEutranMeaGrp.NetworkingOptionOptB1Thld** parameter), the measurement report of event B1 will be sent.

The gNodeB determines the frequencies to be measured and their priorities based on the following principles:

- The gNodeB determines the frequencies to be included in measurement configurations, based on the available band combinations indicated in the EN-DC capabilities of the UE and the NSA PCC frequency priority specified by the **NRCellEutranNFreq.NsaPccFreqPriority** parameter.
A frequency that meets one of the following conditions cannot be selected as a candidate:
 - The **NRCellEutranNFreq.FreqIntrfFlag** parameter is set to **INTRF** for the frequency.
 - The **NRCellEutranNFreq.FreqRelationship** parameter is set to **OVERLAPPED** for the frequency, and the

NSA_FREQ_CONFLICT_AVOID_SW option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.

- The gNodeB can deliver a maximum of six frequencies for measurement. The frequencies that can be selected as PCC frequencies are delivered preferentially. If the number of such frequencies does not reach six, the gNodeB delivers frequencies that can be selected as SCC frequencies as a supplement. The gNodeB arranges the PCC frequencies in descending order of their NSA PCC frequency priorities and arranges the SCC frequencies in descending order of the NSA PCC frequency priorities for the corresponding PCC frequencies. If multiple frequencies have the same priority, the frequencies with the smallest absolute radio frequency channel numbers are preferentially selected.
3. The gNodeB takes different actions, depending on whether it has received measurement reports on inter-RAT event B1 before the handover waiting timer expires.
- If the gNodeB receives measurement reports on inter-RAT event B1 for all frequencies before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations immediately and goes to the next step.
 - If the gNodeB receives measurement reports on inter-RAT event B1 for some of the frequencies before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations after the expiry and goes to the next step.
 - If the gNodeB does not receive measurement reports on any frequency before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations. The procedure ends.
4. (Optional) The gNodeB initiates ANR when no relationship exists with the neighboring LTE cell that has both the highest anchor priority and highest RSRP.

To support handovers from SA to the UE-supported but missed neighboring LTE cell with the highest anchor priority (specified by **NRCCellEutranNFreq.NsaPccFreqPriority**) and highest RSRP after the gNodeB has received all LTE B1 measurement reports, the **NSA_SA_SEL_MISS_CELL_ANR_SW** option of the **NRCCellAnr.AnrAlgoSwitch** parameter can be selected to enable delivery of CGI measurement configurations. The gNodeB then determines whether to add missed neighbor relationships based on the CGI measurement results according to the configured ANR mode. Specifically:

- In ANR controlled mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCCellAnr.AnrModeSwitch** parameter selected):
 - If handovers are allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCCellAnr.AnrModeSwitch** parameter selected): CGI measurement configuration can be delivered. The neighboring LTE cell is added as a to-be-confirmed neighboring cell based on the CGI measurement result, and SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained in 6.
 - If handovers are not allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCCellAnr.AnrModeSwitch** parameter deselected): Handovers to

the to-be-confirmed neighboring cell is not allowed, but CGI measurement configurations are still delivered once. After this neighboring cell is confirmed as a formal one on the OSS, handovers to this neighboring cell can be performed.

- In ANR free mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter deselected): CGI measurement configurations can be delivered. After receiving a CGI measurement report, the gNodeB adds the neighboring cell as a formal one based on the CGI measurement result. SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained in 6.

If the **NSA_SA_SEL_MISS_CELL_ANR_SW** option is deselected, step 5 is performed after step 3 is complete.

5. The gNodeB selects LTE frequencies for resource evaluation, and the eNodeB assists in feeding back the evaluation results.
 - The gNodeB preferentially selects the eNodeB serving the cell with the highest RSRP value on the frequency with the highest NSA PCC frequency priority. Then, the gNodeB checks whether an X2 interface is available between itself and this eNodeB.
 - If an X2 interface is available between the gNodeB and this eNodeB, the gNodeB sends an EN-DC Private Evaluation Request message over the X2 interface, which is an available resource evaluation assistance message used for the evaluation result of available NSA bandwidth resources.
 - If an X2 interface is unavailable between the gNodeB and this eNodeB, the gNodeB selects the eNodeB serving the cell with the highest RSRP value on the frequency with the second highest NSA PCC frequency priority. Then, the gNodeB checks whether an X2 interface is available between itself and this eNodeB. The gNodeB repeats the selection activities until an appropriate eNodeB is found. If no qualified eNodeB is found, the procedure ends.

If there are multiple frequencies with the same NSA PCC frequency priority, the eNodeB serving the cell with the highest RSRP value on the frequency with the smallest absolute radio frequency channel number is selected for the following steps.

- After receiving the available resource evaluation assistance message, the eNodeB evaluates the available NSA bandwidth resources and returns the evaluation result in an EN-DC Private Evaluation Response message.

If CA is working in CA-group-based mode on the eNodeB, the eNodeB considers the PCC resources when evaluating available bandwidth resources. For details about the CA-group-based configuration mode, see *Carrier Aggregation* in 5G RAN Feature Documentation.

If the band combinations include licensed assisted access (LAA) bands, the eNodeB does not consider the resources of LAA SCells when evaluating available bandwidth resources. For details about LAA, see *Licensed Assisted Access (TDD)* in *eRAN Feature Documentation*.

6. The gNodeB compares the evaluation result of available NSA bandwidth resources returned by the eNodeB with the amount of available SA bandwidth

resources. The available NSA or SA bandwidth resources are evaluated based on the following factors:

- Equivalent downlink cell bandwidth: This bandwidth is calculated based on factors such as the downlink cell bandwidth, UE-supported downlink bandwidths, slot configuration, and whether the SCC supports SRS carrier switching.
- NR spectral efficiency: The NR spectral efficiency is approximately converted based on the downlink experience coefficient of NR cells, which is specified by the **NRCellMobilityConfig.NrCellDlExperienceCoeff** parameter.
- Downlink load factor: This factor is calculated based on the number of UEs to be scheduled in the downlink per second.
- Inter-gNodeB CA: Inter-gNodeB CA UEs experience an inter-gNodeB transmission delay, which affects uplink and downlink service experience of these UEs. The impacts can be mitigated by setting the **NRCellCaMgmtConfig.InterGnbExpEvalCoeff** parameter, which specifies the inter-gNodeB experience evaluation coefficient for individual cells.

If user experience barely improves after a handover, the handover is useless. To avoid such handovers, the gNodeB performs the following:

- If (NSA available bandwidth resources – SA available bandwidth resources)/SA available bandwidth resources > **NRInterRatHoParam.DlResourceRatio**, go to the next step. If there are no SA available bandwidth resources due to the **NRCellMobilityConfig.NrCellDlExperienceCoeff** parameter being set to a small value or other reasons, enable NSA/SA downlink selection handover optimization (by selecting the **NSA_SA_DL_SEL_HO_OPT_SW** option of the **NRCellMobilityConfig.InterRatHoExtSwitch** parameter). This enables NSA/SA downlink selection to take effect in such scenarios.
- If not, the entire procedure ends.

7. The UE is handed over to the target LTE cell.

The gNodeB instructs the UE to initiate a handover to the target LTE cell and sends the information about the target LTE cell and the serving NR cell to the eNodeB through a handover request. The serving NR cell is used as the PSCell to be added for the target LTE cell.

Then the UE is handed over to the target LTE cell. In this case, the incoming handover is caused by SA-to-NSA switching based on user experience, the following operations are performed:

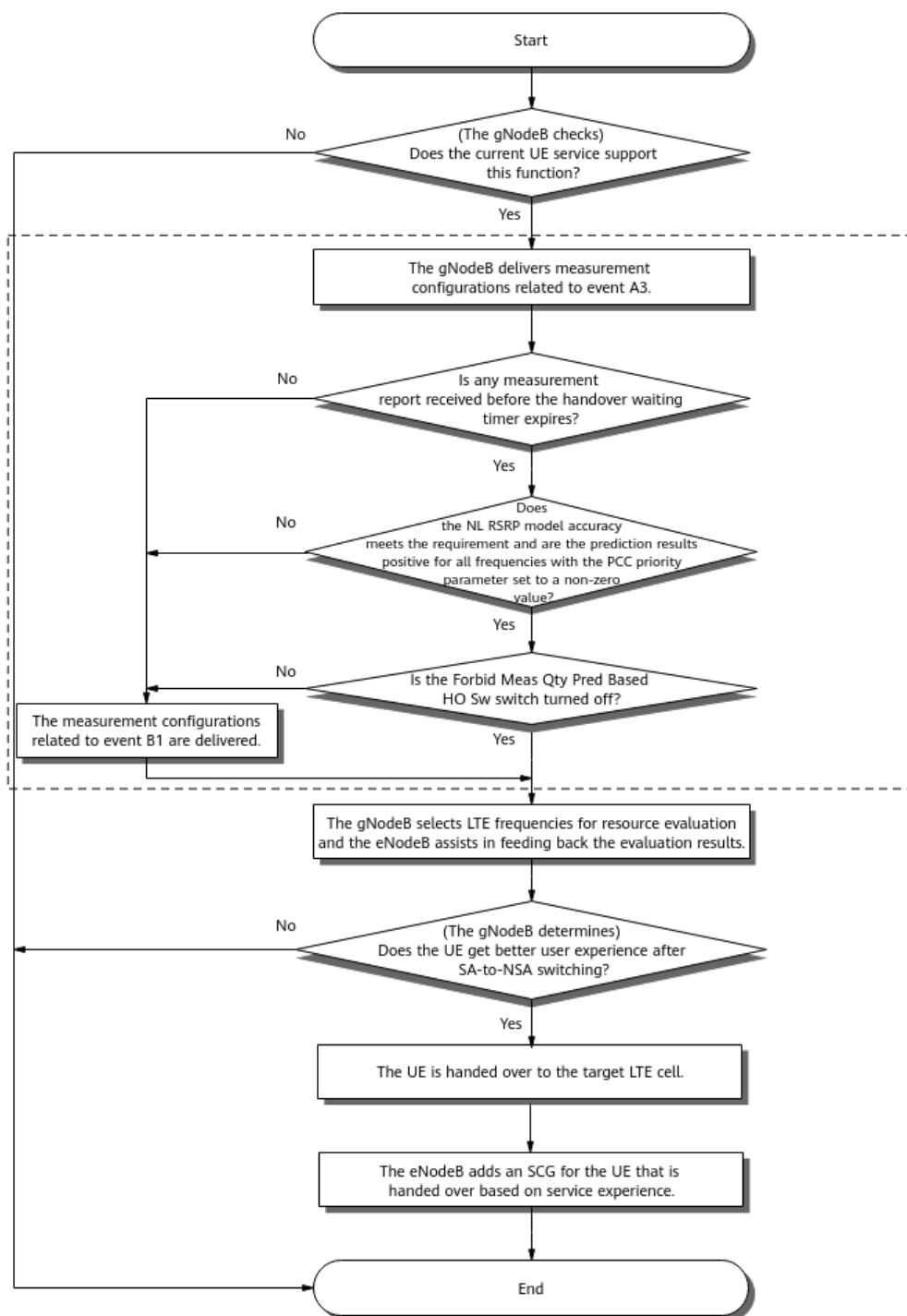
- If the UE supports direct SA-to-NSA switching (supporting the **nr-HO-ToEN-DC-r16** capability specified in 3GPP Release16), and the **DIRECT_NR_HO_TO_EN_DC_SW** option of the eNodeB parameter **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** is selected, the eNodeB directly adds an SCG based on the LTE and NR cell information included in the handover request, enabling the UE to switch from the NR to EN-DC state. For details, see *EPC-based NSA Performance Enhancement*.
- Otherwise, an SCG can be added to set up EN-DC services only after the UE is handed over to the LTE cell.

4.1.1.3 SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids

During NSA/SA selection based on downlink traffic volume, the RSRP predicted by the NL RSRP model can be used to replace the RSRP measured by the UE for handover decision making. This function is controlled by the NSA_SA_SEL_OPT_RSRP_PRED_SW option of the NRCellAlgoSwitch.*InterRatServiceMobilitySw* parameter. (For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.)

[Figure 4-3](#) shows the procedure for SA-to-NSA switching with RSRP predicted based on virtual grids.

Figure 4-3 Procedure for SA-to-NSA switching with RSRP predicted based on virtual grids



Note: The content in the dashed-line box in the figure is the difference between the NSA/SA selection with RSRP predicted based on virtual grids and that based on UE measurements.

This function differs from SA-to-NSA switching based on UE measurements, only in the delivery of event measurement configurations by the gNodeB, the obtaining of RSRP, and timer processing:

1. During SA-to-NSA switching with RSRP predicted based on virtual grids, the gNodeB delivers **A3 measurement configurations** related to the frequency on which a UE camps. If the UE reports event A3 within 1s (that is, before the handover waiting timer expires), the gNodeB uses the measurement result reported (that is, the RSRP values of the serving cell and intra-frequency neighboring cells) to determine the coverage status of candidate LTE frequencies.
2. The RSRP for a neighboring frequency can be predicted using the NL RSRP model, without the need for UE measurements, when all the following conditions are met:
 - The accuracy of the NL RSRP model on the gNodeB meets the accuracy requirement. Whether the model accuracy meets the requirement is determined as follows:
 - When the model accuracy is greater than or equal to the **NRCellSmartMultiCarr.PredAccuForDataThld** parameter value, the accuracy requirement is met and the grid model can be used for data service mobility procedures.
 - When the model accuracy is lower than the **NRCellSmartMultiCarr.PredAccuForDataThld** parameter value, the accuracy requirement is not met and the grid model cannot be used for data service mobility procedures.
 - For all LTE frequencies with the **NRCellEutranNFreq.NsaPccFreqPriority** parameter set to non-zero values, the results of prediction of using the NL RSRP model are either positive or negative.
 - The **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr.HoProtectAlgoSwitch** parameter is deselected.
3. During SA-to-NSA switching with RSRP predicted based on virtual grids, the gNodeB delivers **B1 measurement configurations** to UEs if any of the following conditions is met:
 - The UE does not report event A3 within 1s.
 - The **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr.HoProtectAlgoSwitch** parameter is selected.
 - Among all LTE frequencies with the **NRCellEutranNFreq.NsaPccFreqPriority** parameter set to non-zero values, the results of prediction using the NL RSRP model are not clear, that is, neither positive nor negative, for at least one of these frequencies.

4.1.1.4 Handling After UE Access in NSA

After a UE accesses an NSA network, E-UTRAN to NG-RAN traffic steering or fast return from E-UTRAN to NG-RAN allows the UE to quickly return to an SA network. For details about E-UTRAN to NG-RAN traffic steering and fast return from E-UTRAN to NG-RAN, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

- If flexible user steering is not enabled, the UE can return to SA networking through traffic steering or fast return when the **ServicelrHoCfgGroup.InterRatHoState** parameter is set to **MUST_HO** for at least one QCI among the QCIs of services running on the UE.

- If flexible user steering is also enabled:
 - When the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidoAlgoSwitch** parameter is deselected for the UE, the traffic steering or fast return procedure can be performed if no **Inter-RAT handover state** is set to **NO_HO** for all of the QCI of services running on the UE.
 - When the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidoAlgoSwitch** parameter is selected for the UE, the traffic steering or fast return procedure can be performed if the **Inter-RAT handover state** is set to **MUST_HO** for at least one QCI among the QCIs of services running on the UE.

If the UE is not expected to return to an SA network for anti-ping-pong consideration, the following parameter setting policies are recommended:

- If flexible user steering is not enabled, **Inter-RAT handover state** cannot be set to **MUST_HO** for the QCIs of services running on the UE.
- If flexible user steering is also enabled:
 - If the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidoAlgoSwitch** parameter is deselected for the UE, the **Inter-RAT handover state** must be set to **NO_HO** for at least one QCI among the QCIs of services running on the UE.
 - If the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidoAlgoSwitch** parameter is selected for the UE, the **Inter-RAT handover state** cannot be set to **MUST_HO** for any QCI of services running on the UE.

4.1.1.5 Handling of UEs in Abnormal Scenarios

When some UEs stay in the CA state for a long time on an SA network, abnormal CQI reporting may occur or the bit error ratio (BER) may be high in their SCells due to overheating protection or UE power saving. As a result, user experience of these UEs on the SA network is worse than that in an NSA network. The gNodeB can use the following functions to detect abnormal SCC CQI reporting and a high SCC BER of such UEs:

- Detection of abnormal SCC CQI reported by CA UEs, which is controlled by the **CA_SCC_CQI_ABNORMAL_DETECT_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter
- SCC high BER detection, which is controlled by the **CA_SCC_HIGH_BER_DETECT_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter

When the gNodeB detects that the SCC CQI reported by a CA UE is abnormal or the SCC BER is high, whether to immediately trigger NSA/SA selection based on user experience depends on whether the function NSA/SA selection triggered by abnormal SCCs takes effect. This function is controlled by the **NSA_SA_SEL_SCC_ABN_TRIG_SW** option of the **NRCelAlgoSwitch.MultiFreqAlgoSwitch** parameter.

- If this option is selected, NSA/SA selection based on user experience is immediately triggered in the preceding scenario.
- If this option is deselected, NSA/SA selection based on user experience is not immediately triggered in the preceding scenario.

The preceding functions can work together as follows:

- When both NSA/SA selection triggered by abnormal SCCs and detection of abnormal SCC CQI reported by CA UEs are enabled, if the gNodeB detects that the SCC CQI reported by a high-traffic CA UE is abnormal on an SA network, NSA/SA selection based on downlink traffic volume is triggered. For details about high-traffic UE identification, see [4.1.1.1 gNodeB Identifying High-Traffic UEs](#).
- When both NSA/SA selection triggered by abnormal SCCs and SCC high BER detection are enabled, the gNodeB triggers NSA/SA selection based on downlink traffic volume if the gNodeB detects that the SCC BER of a high-traffic CA UE is high on an SA network. For details about high-traffic UE identification, see [4.1.1.1 gNodeB Identifying High-Traffic UEs](#).
- When all the three functions are enabled, the gNodeB triggers NSA/SA selection based on downlink traffic volume if the gNodeB detects that the SCC CQI reported by a high-traffic CA UE is abnormal or that the SCC BER of a high-traffic CA UE is high on an SA network. For details about high-traffic UE identification, see [4.1.1.1 gNodeB Identifying High-Traffic UEs](#).

4.1.2 Network Analysis

4.1.2.1 Benefits

NSA/SA selection based on downlink traffic volume is recommended in small-bandwidth NR cells. After this function takes effect, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) increases. This function is not recommended for large-bandwidth NR cells.

Scenarios or factors that affect the gains of this function:

- The smaller the NR cell bandwidth and the larger the available LTE bandwidth in NSA networking, the higher the gains of this function.
- In certain scenarios (for example, when a large number of UEs perform handovers to an LTE cell at the same time), the actual available resources are fewer than the estimated available resources. Therefore, the improvement in the average downlink UE throughput after SA-to-NSA switching is insignificant.

NOTE

Small-bandwidth cells:

- NR FDD cells with 20 MHz or lower bandwidth
- NR TDD cells with 40 MHz or lower bandwidth

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

NR side:

- The downlink RB usage of NR cells, indicated by the **Downlink Resource Block Utilizing Rate (DU)**, decreases.
- The downlink traffic volume in NR cells, indicated by the **Downlink Traffic Volume (DU)**, decreases.

- During switching from SA to NSA, for a period of time, the average downlink UE throughput, indicated by the **User Downlink Average Throughput (DU)** counter, decreases.
- After a UE switches from SA to NSA, the UE transmit power decreases, affecting the average uplink UE throughput, indicated by the **User Uplink Average Throughput (DU)** counter, when it is located at the cell edge. The loss in the average uplink UE throughput can almost be offset by using dynamic power sharing if the UE supports the sharing function.
- The proportion of UEs camping in SA networking decreases.
- If the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled (by selecting the **UL_TFC_NR_LTE_MOBILITY_ENH_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter), the **N.NsaDc.SgNB.Mod.Req.Att** counter value decreases.
- If detection of abnormal CQI reported by CA UEs and SCC high BER detection are enabled, in those abnormal scenarios, the downlink throughput of the CA UE increases after it is handed over from SA to NSA. In addition, the values of the following counters are expected to increase:
 - **N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut**
 - **N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut**
 - **N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut**
 - **L.IRATHO.N2E.Direct2EnDc.PrepAttIn**
 - **L.IRATHO.N2E.Direct2EnDc.ExecSuccIn**
 - **L.IRATHO.N2E.Direct2EnDc.PrepInSucc**
- **Table 4-1** describes the change trends of the number of SA-to-NSA switching times when specific functions are enabled. **Table 4-2** lists the counters related to the number of SA-to-NSA switching times.

Table 4-1 Change trends of the number of SA-to-NSA switching times when specific functions are enabled

Function Name	Change Trend of the Number of SA-to-NSA Switching Times
NSA/SA selection based on user experience for inter-gNodeB CA UEs	Increase
ANR for missing neighboring LTE cells	Increase
NSA/SA downlink selection handover optimization	Increase

Table 4-2 Counters related to the number of SA-to-NSA switching times

Counter ID	Counter Name
1911829365	N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut

Counter ID	Counter Name
1911829366	N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
1911829367	N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
1911853802	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuc cOut.DL
1911853803	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAtt Out.DL
1911853807	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAtt Out.DL
1526765179	L.IRATHO.N2E.Direct2EnDc.ExecSuccln
1526765175	L.IRATHO.N2E.Direct2EnDc.PrepAttln
1526765176	L.IRATHO.N2E.Direct2EnDc.PrepInSucc
1526749536	L.IRATHO.N2E.ExecSuccln
1526749534	L.IRATHO.N2E.PrepAttln
1526749535	L.IRATHO.N2E.PrepInSucc

LTE side:

- The downlink RB usage of LTE cells, indicated by **Downlink Resource Block Utilizing Rate**, increases.
- The downlink traffic volume in LTE cells, indicated by **Downlink Traffic Volume**, increases.
- The downlink RB usage of NSA DC UEs, indicated by the **Downlink Resource Block Utilizing Rate (NSA DC)**, increases.
- When the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled, the **L.NsaDc.SCG.Mod.Req.Att** counter value decreases.
- The average uplink or downlink UE throughputs decrease if LTE cell quality deteriorates or the number of available RBs decreases. The specific scenarios are exemplified as follows:
 - A large number of UEs are handed over to an LTE cell. In this scenario, the number of available RBs is less than the estimated number.
 - After a UE switches from NR TDD SA to NSA, the uplink transmit capability of the UE is degraded, rolled back from 2T to 1T, which causes the average uplink UE throughput to decrease in certain scenarios.

4.1.3 Requirements

4.1.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	MRFD-171222	NSA/SA Selection Based on User Experience (LTE FDD)	LT1S0NSA SA00	NSA/SA Selection Based on User Experience (LTE FDD)	eNodeB	per eNodeB
LTE TDD	MRFD-171232	NSA/SA Selection Based on User Experience (LTE TDD)	LT4SCSEH QTDD	NSA/SA Selection Based on User Experience (LTE TDD)	eNodeB	per eNodeB
NR	MRFD-171262	NSA/SA Selection Based on User Experience (NR)	NR0S0NS BUE00	NSA/SA Selection Based on User Experience (NR)	gNodeB	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD	Downlink dynamic data split	gNBpdcParamGroup.DlDataPdcSplitMode set to SCG_AND_MCG	<i>EPC-based NSA Basics</i>
NR FDD Low-frequency NR TDD	NSA carrier combination selection based on user experience ^a	NSA_LTE_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>EPC-based NSA Experience Improvement</i>
NR FDD Low-frequency NR TDD	NSA/SA selection based on downlink traffic volume ^a	NSA_SA_DL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
NR FDD Low-frequency NR TDD	NSA/SA selection based on uplink coverage ^a	NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
NR FDD Low-frequency NR TDD	NL RSRP model building ^b	LNR_HO_MEAS_QTY_PRED_FUNCTION option of the NRCellSmartMultiCarrierLnrMultiCarrierAlgorithm parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>
NR FDD Low-frequency NR TDD	NSA/SA selection based on downlink traffic volume ^b	NSA_SA_DL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>

RAT	Function Name	Function Switch	Reference
Low-frequency NR TDD	NSA/SA selection based on uplink coverage ^b	NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination ^b	VG_MODEL_ALL_OW_BUILD_FLAG option of the NRCeEutranNfReq.AggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i>
NR FDD Low-frequency NR TDD	Inter-RAT ANR ^c	NR_EUTRAN_ANR_SW option of the NRCeAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>
Note: a: If the NSA_SA_SEL_INTER_GNB_CA_SW option of the gNodeBParam.NetworkingOptionOptSw parameter is selected, the NSA_LTE_SEL_OPT_SW, NSA_SA_DL_SEL_OPT_SW, or NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter must be selected. b: SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids requires that (1) NL RSRP model building be enabled, (2) either NSA/SA selection based on downlink traffic volume or NSA/SA selection based on uplink coverage be enabled, and (3) Virtual Grid-based Multi-Frequency Coordination be enabled for at least one neighboring LTE frequency of the NR cell. c: The NSA_SA_SEL_MISS_CELL_ANR_SW option can be selected only when the NR_EUTRAN_ANR_SW option is selected.			

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Downlink CA	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	The gNodeB carries the selected EN-DC combination in the selectedbandCombinationInfoEN-DC-v1540 IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA networking. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on downlink traffic volume is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.
LTE FDD LTE TDD	E-UTRAN to NG-RAN traffic steering	ServBasedNrHoSwitch option of the CellAlgoSwitch.HoAllowedSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	After NSA/SA selection based on downlink traffic volume is enabled, the evaluation of E-UTRAN to NG-RAN traffic steering and that of fast return from E-UTRAN to NG-RAN are changed. This results in a lower probability that E-UTRAN to NG-RAN traffic steering and fast return from E-UTRAN to NG-RAN take effect, and therefore smaller values of the L.IRATHO.E2N.Service.PrepareAttOut and L.IRATHO.E2N.FastReturn.PrepareAttOut counters.
LTE FDD LTE TDD	Fast return from E-UTRAN to NG-RAN	FAST_RETURN_TO_NR_SW option of the CellAlgoExtSwitch.HoAllowedSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Operator-specific RB management	• RB_DYNAMIC_SHARING_SWITCH option of the NRDU CellAI goSwitch.RanSharingAlgorithm parameter • MAX_RB_LIMIT_SWITCH option of the NRDU CellAI goSwitch.RanSharingAlgorithm parameter • MIN_RB_GUARANTEE_SWITCH option of the NRDU CellAI goSwitch.RanSharingAlgorithm parameter	<i>Multi-Operator Sharing in 5G RAN Feature Documentation</i>	In RAN sharing with common carrier scenarios, if operator-specific RB management is enabled and the shared cells do not share all their resources, NSA/SA selection based on downlink traffic volume is not recommended. Otherwise, user experience evaluation may be inaccurate, resulting in user experience deterioration.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_C APB_MO B_TO_HP LMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In NSA/SA selection based on downlink traffic volume, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
LTE FDD LTE TDD	FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs	NsaDcAlgoParam.NrTddScsRsrpThld set to a value other than 255	<i>EPC-based NSA Basics</i>	When NSA/SA selection based on downlink traffic volume takes effect, the function of FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs does not take effect.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all boards that support NSA networking support this function. For details, see *EPC-based NSA Basics*.

- For NR, all NR-capable boards support this function.

RF Modules

No requirements

4.1.3.4 Networking

The NSA networking must use the Option 3x architecture.

4.1.3.5 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later and support the following capabilities:
 - SA capability
 - **handoverLTE-EPC** capability
 - EN-DC capability (whether the UE whose **RAT-Type** is "eutra" supports "en-DC-r15" capability); and the *supportedBandListEN-DC-r15* IE in the UE capability including the band to which the current NR cell belongs
 - **nr-HO-ToEN-DC-r16** capability specified in 3GPP Release 16, if SA-to-NSA switching is required
- Core network requirements
 - The core network must comply with 3GPP Release 15 or later.
 - The N26 interface must be already configured between the MME and AMF.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-3, **Table 4-4**, and **Table 4-5** describe the parameters related to NSA/SA selection based on downlink traffic volume. **Table 4-6** and **Table 4-7** list the recommended configurations for various services/UEs. For details about the basic parameters for interoperability from NG-RAN to E-UTRAN, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* > "Data Preparation" and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* > "Basic Mobility Management Procedure for UEs in Connected Mode" > "Operation and Maintenance (NG-RAN)" > "Data Preparation."

Table 4-3 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Service Mobility Switch	NRCeAlgoSwitc h.InterRatService MobilitySw	Select the MOBILITY_TO_EUTRAN_SW option.

Parameter Name	Parameter ID	Setting Notes
Networking Option Opt Sw	gNodeBParam.N etworkingOption OptSw	Select the NSA_SA_DL_SEL_OPT_SW option.
Operator Inter-RAT Policy Switch	gNBOperator.Op eratorInterRatPol icySw	Select the NSA_SA_DL_SEL_OPT_SW option.
Networking Option Opt Flag	gNBQciBearer.Ne tworkingOption OptFlag	It is recommended that the parameter settings for different QCLs on the gNodeB be consistent with those for the corresponding QCLs on the eNodeB. The parameter on the eNodeB side is ServiceIrHoCfgGroup.InterRatHoState . For details about the recommended settings, see Table 4-6 .
Networking Option Opt Flag	gNBRfspConfig.N etworkingOption OptFlag	This parameter must be set if flexible user steering is enabled and different network architecture selection policies are required for UEs with different RFSP indexes. It is recommended that the policies configured for different UE types on the gNodeB be consistent with those configured for the corresponding UE types on the eNodeB. The selection policy on the eNodeB side is controlled by the NSA_SA_SEL_OPT_SW option of the SpidCfg.SpidoAlgoSwitch parameter. For details about the recommended settings, see Table 4-7 .
Inter-RAT Service Mobility Switch	NRCelAlgoSwitch h.InterRatService MobilitySw	To use virtual grids for NSA/SA selection based on user experience, select the NSA_SA_SEL_OPT_RSRP_PRED_SW option.
Networking Option Opt Sw	gNodeBParam.N etworkingOption OptSw	It is recommended that the NSA_SA_SEL_INTER_GNB_CA_SW option be selected in inter-gNodeB CA scenarios.

Table 4-4 NR parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
NSA/SA Selection Opt Period	gNodeBParam.NsaSaSelOptPeriod	The default value is recommended.
DL Resource Ratio	NRInterRatHoParam.DLResourceRatio	The default value is recommended. If the value of this parameter is too small, the number of RBs available for LTE decreases, which may cause negative gains.
NSA/SA Selection Opt High Traffic Thld	NRDUCellCarrMgmt.NsaSaSelOptHighTrafficThld	The default value is recommended. If the value of this parameter is too small and a large number of SA UEs are switched to NSA, resources in the LTE cell decrease and NSA DC gains cannot be obtained.
NSA/SA Selection Opt Low Traffic Thld	NRDUCellCarrMgmt.NsaSaSelOptLowTrafficThld	The default value is recommended.
NSA PCC Frequency Priority	NRCeUtranNFrq.NsaPccFreqPriority	The value of this parameter must be the same as that of the eNodeB parameter PccFreqCfg.NsaPccAnchoringPriority .
ANR Algorithm Switch	NRCeUtran.NrAnrAlgorithmSwitch	It is recommended that the NSA_SA_SEL_MISS_CELL_ANR_SW option be selected when UEs need to be handed over from SA to the neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor relationships with that cell.
ANR Mode Switch	NRCeUtran.NrAnrModeSwitch	Set this parameter based on the network plan when UEs need to be handed over from SA to the neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor relationships with that cell.
Networking Option Opt RSRP Threshold	NRCeUtranMeaGrp.NetworkingOptionOptB1Thld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Multi-Frequency Algorithm Switch	NRCellAlgoSwitch <i>h.MultiFreqAlgoSwitch</i>	If a UE stays in the CA state for a long time on an SA network, abnormal CQI reporting may occur or the BER may be high in their SCells due to overheating protection or UE power saving. In this case, it is recommended that the NSA_SA_SEL_SCC_ABN_TRIG_SW option be selected.
Multi-Carrier Switch	NRDUCellFeatureSw <i>MultiCarrierSwitch</i>	If a UE stays in the CA state for a long time on an SA network, abnormal CQI reporting may occur in the SCell due to overheating protection or UE power saving. In this case, it is recommended that the CA_SCC_CQI_ABNORMAL_DETECT_SW option be selected.
Multi-Carrier Switch	NRDUCellFeatureSw <i>MultiCarrierSwitch</i>	If UEs stay in the CA state for a long time on an SA network, a high BER may occur in the SCell due to overheating protection or UE power saving. In this case, it is recommended that the CA_SCC_HIGH_BER_DETECT_SW option be selected.
Inter-gNodeB Exp Evaluation Coefficient	NRCellCaMgmtConfig <i>InterGnbExpEvalCoeff</i>	Set this parameter based on the network plan.
Inter-RAT Handover Extend Switch	NRCellMobilityConfig <i>InterRatHoExtSwitch</i>	Select the NSA_SA_DL_SEL_HO_OPT_SW option if NSA/SA downlink selection is required but there are no available SA bandwidth resources for UEs.
Mobility HO Control Switch	NRInterRatHoParam <i>MobilityHOControlSwitch</i>	If NSA/SA selection based on downlink traffic volume is required for a cell, deselect the NSA_SA_DL_OPT_FRBD_SW option for this cell.

Parameter Name	Parameter ID	Setting Notes
Process Switch	NRCellAlgoSwitch.<i>ProcessSwitch</i>	It is recommended that the UE_CAP_SEPARATE_QUERY_SW option be selected if the gNodeB is required to query the NR, EUTRA-NR, and EUTRA capabilities of UEs separately. If UE capability separate query is not required for specific UEs, select the UE_CAPB_SEPARATE_QUERY_SW_OFF option of the gNBUECompatSw.<i>BlacklistCtrlSwitch</i> parameter.

Table 4-5 LTE parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	EnodebAlgoExtSwitch.<i>MultiNetworkingOptionOptSw</i>	Select the LTE_FDD_NSA_SA_DL_SEL_OPT_SW or LTE_TDD_NSA_SA_DL_SEL_OPT_SW option.
Handover to NR Algo Switch	EnodebAlgoExtSwitch.<i>NrHandoverAlgoSwitch</i>	Select the DIRECT_NR_HO_TO_EN_DC_SW option.
SPID Algorithm Switch	SpidCfg.<i>SpidAlgoSwitch</i>	This parameter must be set if flexible user steering is enabled and different network architecture selection policies are required for UEs with different SPIDs. Select the NSA_SA_SEL_OPT_SW option for the SPID of UEs that need to support the network architecture selection function.

Table 4-6 Recommended settings for different services

Typical Services and Their Policies	Recommended Value of <i>gNBQciBearer.NetworkingOptionOptFlag</i> on the gNodeB	Recommended Value of <i>ServiceIrHoCfgGroup.InterRatHoState</i> on the eNodeB
Enhanced Mobile Broadband (eMBB) services, such as QCI 9 services: Data split in NSA is recommended. SA-to-NSA switching based on user experience is allowed.	PERMIT	PERMIT_HO
Ultra-reliable low-latency communication (URLLC) services: They can be carried only in NR SA networks. Therefore, when URLLC services are initiated, UEs must return to NR SA networks as soon as possible.	FORBID	MUST_HO
Voice services, such as QCI 1 services: Set the policy as required.	FORBID	NO_HO is recommended if it is expected to reduce the number of handovers for voice services. PERMIT_HO or MUST_HO is recommended if VoNR is preferred.
Wireless broadband (WBB) services: Mobility reduction is recommended.	FORBID	PERMIT_HO
Other types of services, such as QCI 5 services: Use the follow-up policy.	FOLLOW	PERMIT_HO

Table 4-7 Recommended settings for different types of UEs

UE Type	Recommended Value of <i>gNBRfspConfig.NetworkingOptionOptFlag</i> on the gNodeB	Recommended Value of <i>SpidCfg.SpidAlgoSwitch</i> on the eNodeB
Common UEs	TRUE	NSA_SA_SEL_OPT_SW selected
Special UEs	FALSE	NSA_SA_SEL_OPT_SW deselected

4.1.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must be already completed before activating this function. For details, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* > "Using MML Commands" and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* > "Basic Mobility Management Procedure for UEs in Connected Mode" > "Operation and Maintenance (NG-RAN)" > "Using MML Commands".

Activation Command Examples

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//On the gNodeB side
//Turning on the switch of mobility to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1;
//Turning on the NSA/SA selection optimization switch
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_DL_SEL_OPT_SW-1, NsaSaSelOptPeriod=60;
ADD GNBOPERATOR: OperatorId=1, OperatorName="PRIMARY_OPERATOR", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_DL_SEL_OPT_SW-1;
//Deselecting the NSA_SA_DL_OPT_FRBD_SW option for a target cell to enable NSA/SA selection based on
downlink traffic volume for this cell
MOD NRINTERRATHOPARAM: NrCellId=0, MobilityHOControlSwitch=NSA_SA_DL_OPT_FRBD_SW-0;
//(Optional) Setting NetworkingOptionOptFlag if network architecture selection policies need to be
configured for different QCI
MOD GNBQCI BEARER: Qci=1, NetworkingOptionOptFlag=FORBID;
MOD GNBQCI BEARER: Qci=9, NetworkingOptionOptFlag=PERMIT;
//(Optional; required only when RSRP prediction based on virtual grids is to be used) Enabling RSRP
prediction for NSA/SA selection optimization
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=NSA_SA_SEL_OPT_RSRP_PRED_SW-1;
//(Optional; required only when flexible user steering is enabled) Setting the network architecture selection
policy for an RFSP index
MOD GNBFRFSPCONFIG: OperatorId=1, RfspIndex=1, NetworkingOptionOptFlag=TRUE;
//(Optional, required only in inter-gNodeB CA scenarios) Turning on the NSA/SA Sel for Inter-gNodeB CA
```

```

Switch
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-1;
//(Optional, required only in inter-gNodeB CA scenarios) Setting the coefficient used for experience
evaluation in inter-gNodeB scenarios
MOD NRCELLCAMGMTCONFIG: NrCellId=0, InterGnbExpEvalCoeff=70;

//On the eNodeB side
//Turning on the NSA/SA selection optimization switch
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_DL_SEL_OPT_SW-1&LTE_TDD_NSA_SA_DL_SEL_OPT_SW-1,
NrHandoverAlgoSwitch=DIRECT_NR_HO_TO_EN_DC_SW-1;
//(Optional) Setting service-based inter-RAT handover policies if network architecture selection policies
need to be configured for different QCI (The policies must be consistent with those configured on the
gNodeB. Otherwise, a ping-pong effect will occur.)
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=1, ServiceHoNrPolicyGroupId=0;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, ServiceHoNrPolicyGroupId=1;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=0, InterRatHoState=NO_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=1, InterRatHoState=PERMIT_HO;
//(Optional; required only when flexible user steering is enabled) Setting the network architecture selection
policy for an SPID. The policy must be consistent with that configured on the gNodeB.
MOD SPIDCFG: Spid=1, SpidAlgoSwitch=NSA_SA_SEL_OPT_SW-1;

```

Optimization Command Examples

```

//On the gNodeB side
//Setting the downlink resource ratio
MOD NRINTERRATHOPARAM: NrCellId=0, DlResourceRatio=10;
//Setting the thresholds for determining the high and low traffic states
MOD NRDUCELLCARRMGMT: NrDuCellId=0, NsaSaSelOptHighTrafficThld=10, NsaSaSelOptLowTrafficThld=2;
//Setting the NSA PCC frequency priority
MOD NRCELLEUTRANNFREQ: NrCellId=0, DlEarfcn=1300, NsaPccFreqPriority=10;
//(Optional) Run the following commands if UEs are required to be handed over from SA to the
neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor
relationships with this neighboring cell
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NSA_SA_SEL_MISS_CELL_ANR_SW-1;
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_EUTRAN_ANR_CTRL_MODE_SW-0;
//Setting the RSRP threshold for event B1 related to network architecture selection
MOD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
NetworkingOptionOptB1Thld=-111;
//Turning on the following switches to prevent abnormal SCC CQI reporting or a high SCC BER from
occurring due to overheating protection or UE power saving when a UE stays in the CA state for a long
time on an SA network
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=NSA_SA_SEL_SCC_ABN_TRIG_SW-1;
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=CA_SCC_CQI_ABNORMAL_DETECT_SW-1;
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=CA_SCC_HIGH_BER_DETECT_SW-1;
//(Optional, recommended) Selecting the NSA_SA_DL_SEL_HO_OPT_SW option when there are no available
SA bandwidth resources for UEs
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterRatHoExtSwitch=NSA_SA_DL_SEL_HO_OPT_SW-1;
//(Optional) Selecting the UE_CAP_SEPARATE_QUERY_SW option if the gNodeB is required to query the NR,
EUTRA-NR, and EUTRA capabilities of UEs separately
MOD NRCELLALGOSWITCH: NrCellId=0, ProcessSwitch=UE_CAP_SEPARATE_QUERY_SW-1;
//(Optional) Selecting the UE_CAPB_SEPARATE_QUERY_SW_OFF option when UE capability separate query
is not needed for specific UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlSwitch=UE_CAPB_SEPARATE_QUERY_SW_OFF-1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivating NSA/SA selection based on downlink traffic volume for the base station

```

//On the gNodeB side
//(Optional, required only in inter-gNodeB CA scenarios) Turning off the NSA/SA Sel for Inter-gNodeB CA

```

```
Switch
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-0;
//(Optional) Disabling RSRP prediction based on virtual grids
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=NSA_SA_SEL_OPT_RSRP_PRED_SW-0;
//Disabling NSA/SA selection optimization
MOD GNBOperator: OperatorId=1, OperatorName="PRIMARY OPERATOR", Mcc="460", Mnc="00",
OperatorInterRatPolicySw=NSA_SA_DL_SEL_OPT_SW-0;
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_DL_SEL_OPT_SW-0;

//On the eNodeB side
//Disabling NSA/SA selection optimization
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_DL_SEL_OPT_SW-0&LTE_TDD_NSA_SA_DL_SEL_OPT_SW-0;
```

Deactivating NSA/SA selection based on downlink traffic volume for a cell

```
//Selecting the NSA_SA_DL_OPT_FRBD_SW option to disable NSA/SA selection based on downlink traffic
volume for the desired cell
MOD NRINTERRATHOPARAM: NrCellId=0, MobilityHOControlSwitch=NSA_SA_DL_OPT_FRBD_SW-1;
```

Deactivating ANR for missing neighboring LTE cells for a cell

```
//Running the following command if UEs are not expected to be handed over from SA to the neighboring
LTE cell with the highest anchor priority and RSRP and the gNodeB does not have neighbor relationships
with this neighboring cell
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NSA_SA_SEL_MISS_CELL_ANR_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

[Using Signaling Tracing](#) and [Using Counters](#) can be used to verify whether NSA/SA selection based on downlink traffic volume has taken effect.

Using Signaling Tracing

To verify whether NSA/SA selection based on downlink traffic volume has been activated, perform the following steps:

- Step 1** Start Uu, NG, and X2 interface tracing for the gNodeB.
- Step 2** Start X2 interface tracing for the eNodeB.
- Step 3** Observe X2 messages on the gNodeB side. If X2 messages include an EN-DC Private Evaluation Request message that does not carry the Evaluation Direction Indication field, the resource evaluation request message for NSA/SA selection based on downlink traffic volume has been successfully sent.
- Step 4** Observe Uu messages on the gNodeB side. If the value of the *targetRAT-Type* IE in the MobilityFromNRCommand message sent by the gNodeB to the relevant UE is "eutra", the preparation for the handover from NG-RAN to E-UTRAN has succeeded.
- Step 5** Observe NG messages on the gNodeB side. If the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the *Successful Handover* IE from the AMF, the handover from NG-RAN to E-UTRAN has been successfully executed.

- Step 6** Observe X2 messages on the eNodeB side. If there is an SgNB Addition Ack message, it indicates that an SCG has been successfully added.

----End

View the **N.DLHighTraffic.UE.Count** counter to obtain the number of times UEs with high traffic volume are identified in a cell.

To verify whether NSA/SA selection with RSRP predicted based on virtual grids has taken effect, check the number of outgoing handover attempts based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback), which equals **N.HO.LnrVirtualGrid.Model.PrepAttOut** minus **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut**. If the number is larger than zero, NSA/SA selection with RSRP predicted based on virtual grids has taken effect.

Using Counters

If any of the counters listed in [Table 4-8](#) is not 0, NSA/SA selection based on downlink traffic volume has taken effect.

Table 4-8 Counters related to NSA/SA selection based on downlink traffic volume

Counter ID	Counter Name
1911853807	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.DL
1911853803	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.DL
1911853802	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.DL

4.1.4.3 Network Monitoring

To evaluate the performance of NSA/SA selection based on UE measurements, monitor the counters listed in [Table 4-9](#).

Table 4-9 Counters used to monitor NSA/SA selection based on UE measurements

Counter ID	Counter Name
1911829365	N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
1911829366	N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
1911829367	N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
1526765175	L.IRATHO.N2E.Direct2EnDc.PrepAttIn
1526765176	L.IRATHO.N2E.Direct2EnDc.PrepInSucc
1526765179	L.IRATHO.N2E.Direct2EnDc.ExecSuccIn

To evaluate the performance of NSA/SA selection with RSRP predicted based on virtual grids, monitor the results of the following formulas:

- Number of successful handovers based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.ExecSuccOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecSuccOut**
- Number of outgoing handover executions based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.ExecAttOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecAttOut**
- Number of outgoing handover attempts based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.PrepAttOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut**

Table 4-10 NR counters used to monitor NSA/SA selection with RSRP predicted based on virtual grids

Counter ID	Counter Name
1911831568	N.HO.LnrVirtualGrid.Model.ExecSuccOut
1911831569	N.HO.LnrVirtualGrid.Model.ExecAttOut
1911831570	N.HO.LnrVirtualGrid.Model.PrepAttOut
1911831561	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecSuccOut
1911831562	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecAttOut
1911831563	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut

4.2 NSA-to-SA Switching Based on Downlink User Experience

NSA-to-SA switching based on downlink user experience is triggered for UEs whose downlink experience in NR cells is considered better than that in LTE cells. This function provides a networking scheme with better service experience for UEs and improves the camping probability of UEs in SA networking.

This function is implemented in two scenarios:

1. NSA-to-SA switching based on downlink user experience is triggered for CEUs that access sub-1 GHz cells under the eNodeB. For details, see [4.2.1.1 Cell Edge Scenario](#).
2. NSA-to-SA switching based on downlink user experience is triggered for UEs that were first handed over to the eNodeB due to SA-to-NSA switching based on downlink traffic volume and then transitioned from the large-packet state to the small-packet state. For details, see [4.2.1.2 Downlink Small-Packet Scenario](#).

NOTE

Sub-1 GHz refers to all frequency bands in which the cell's downlink center frequency is below 1 GHz.

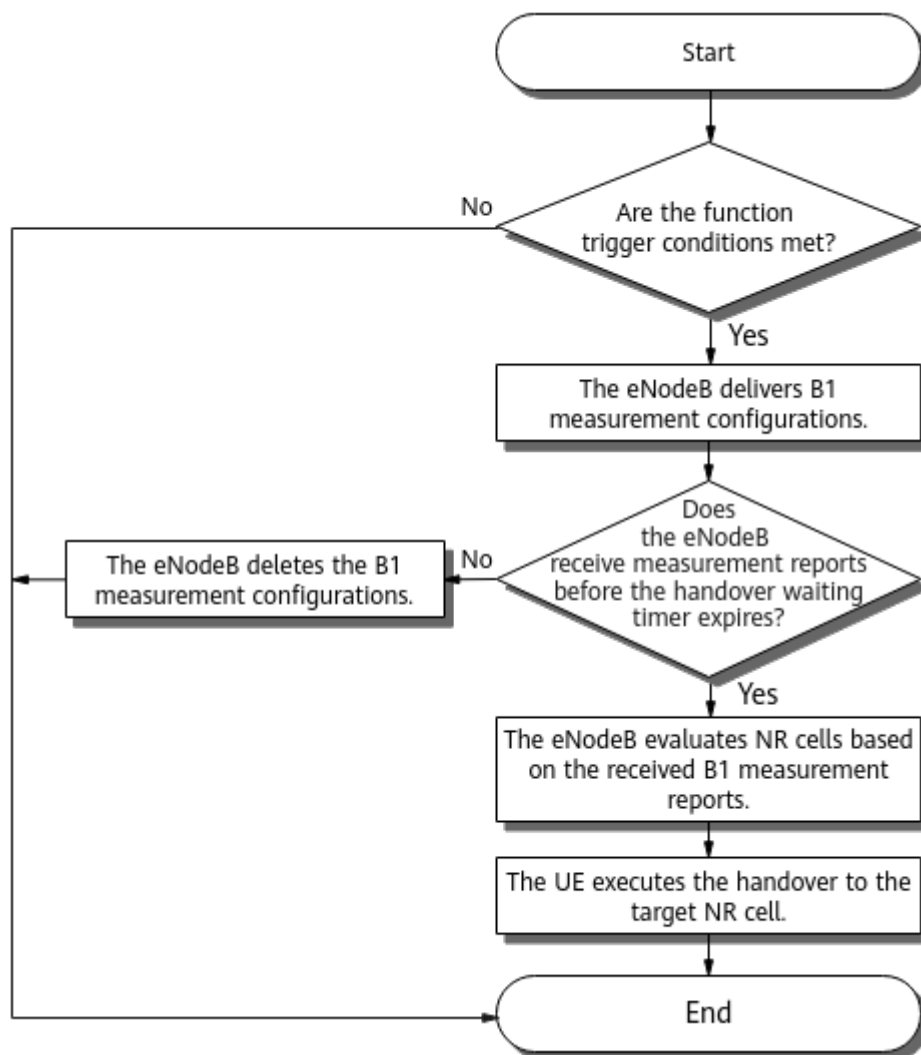
If the conditions for both scenarios are met, scenario 2 preferentially takes effect.

4.2.1 Principles

4.2.1.1 Cell Edge Scenario

If a UE at the cell edge (referred to as a CEU) accesses an LTE sub-1 GHz cell, the eNodeB determines that the UE can have better service experience after being handed over to an NR cell. Therefore, the eNodeB initiates NSA-to-SA switching based on downlink user experience for the UE. [Figure 4-4](#) shows the procedure. This function is controlled by the **DL_NSA_TO_SA_SWITCH_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

Figure 4-4 NSA-to-SA switching based on downlink user experience for a CEU



The procedure is as follows:

1. The eNodeB checks whether the function trigger conditions are met. The conditions for determining whether to trigger the function slightly vary, depending on whether the A2-based SCell removal function is enabled.
 - When A2-based SCell removal is enabled (by selecting the **SccA2RmvSwitch** option of the **ENodeBALgoSwitch.CaAlgoSwitch** parameter), the eNodeB checks whether either of the following conditions is met:
 - The UE accesses the LTE sub-1 GHz cell through initial access, incoming handover, or incoming RRC connection reestablishment.
 - The SCC and SCG of the UE have been removed, and the UE is in an LTE sub-1 GHz cell.

When either of the preceding conditions is met, the eNodeB checks whether the UE is in a carrier selection procedure (such as NSA carrier combination selection based on user experience or Multi-carrier Unified Scheduling). If the UE is not in any other carrier selection procedure, the eNodeB proceeds to step 2. If the UE is in a carrier selection procedure, the eNodeB proceeds to step 2 when the UE stays in the LTE sub-1 GHz cell after the carrier selection procedure is complete.

- When A2-based SCell removal is disabled (by deselecting the **SccA2RmvSwitch** option of the **ENodeBALgoSwitch.CaAlgoSwitch** parameter), the eNodeB checks whether either of the following conditions is met:
 - The UE accesses the LTE sub-1 GHz cell through initial access, incoming handover, or incoming RRC connection reestablishment.
 - The SCG of the UE has been removed, and the PCell of the UE is an LTE sub-1 GHz cell.

When either of the preceding conditions is met, the eNodeB delivers A2 measurement configurations. If the eNodeB receives an A2 measurement report from the UE, the eNodeB checks whether the UE is in a carrier selection procedure. If the UE is not in another carrier selection procedure and the RSRP values of all SCCs of the UE meet the release threshold (jointly determined by **CaMgtCfg.CarrAggrA2ThdRsrp** and **SccFreqCfg.SccA2Offset**), the eNodeB proceeds to step 2. Otherwise, the procedure ends.

2. The eNodeB delivers inter-RAT B1 measurement configurations.

The eNodeB delivers inter-RAT B1 measurement configurations to the UE and starts a handover waiting timer (fixed at 2s). In these configurations, the priorities of the LTE sub-1 GHz cell's neighboring NR cells are arranged in descending order of priority.

The eNodeB determines the frequencies to be included in measurement configurations, based on the available bands indicated in the NR capabilities of the UE and the **NrNFreq.SaPccFreqPriority** parameter setting. A frequency that meets either of the following conditions cannot be selected for delivery of inter-RAT B1 measurement configurations:

- The frequency is not the one used by the LTE sub-1 GHz cell.
- The converted downlink bandwidth of the NR cell operating on the frequency is lower than the downlink bandwidth of the current LTE sub-1

GHz cell. The converted downlink bandwidth of the NR cell equals the NR downlink bandwidth (specified by **NrNFreq.NrDLBandwidth**) multiplied by the NR cell downlink experience coefficient (specified by **NsaDcMgmtConfig.NrCellDLExperienceCoeff**).

3. The eNodeB takes different actions depending on whether inter-RAT B1 measurement reports are received before the handover waiting timer expires.
 - If B1 measurement reports are received on all frequencies before the timer expires, the eNodeB deletes the B1 measurement configurations and proceeds to step 4.
 - If B1 measurement reports are received on some frequencies before the timer expires, the eNodeB deletes the B1 measurement configurations after the timer expires and then proceeds to step 4.
 - If no measurement reports are received on any frequency before the timer expires, the eNodeB deletes the B1 measurement configurations, and the procedure ends.

NOTE

If a B1 measurement report is received on the highest-priority frequency before the timer expires, the eNodeB directly proceeds to step 4 without waiting for measurement reports on other frequencies. It does not delete the B1 measurement configurations when receiving such a report and proceeding to the next step. If the highest-priority frequency is qualified for handover, the eNodeB ends the procedure and performs the handover. If the highest-priority frequency is not qualified for handover, the eNodeB performs step 4 for other frequencies on which measurement reports are received. If no measurement reports are received on other frequencies, the procedure ends.

4. The eNodeB evaluates NR cells based on the received B1 measurement reports.

The eNodeB selects the frequency with the highest SA PCC frequency priority (specified by **NrNFreq.SaPccFreqPriority**) among all NR frequencies on which measurement reports are received, and then selects the cell with the highest RSRP on that frequency as the target NR cell. If multiple cells meet the handover requirements, one of them is randomly selected as the target NR cell.

5. The UE executes the handover.

The eNodeB instructs the UE to initiate a handover to the target NR cell.

4.2.1.2 Downlink Small-Packet Scenario

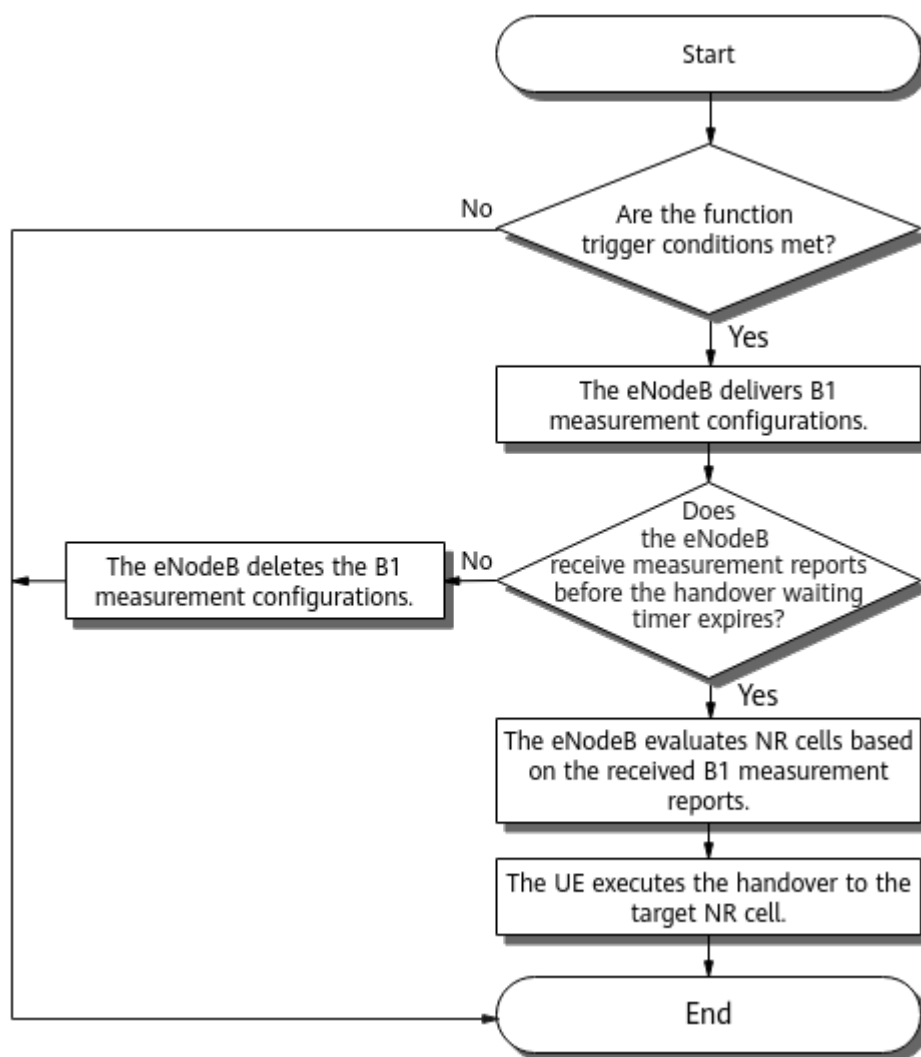
When a UE performing large-packet downlink services is handed over to the eNodeB due to SA-to-NSA switching based on downlink traffic volume and then transitions from the large-packet state to the small-packet state, the eNodeB determines that the UE's experience in the LTE or NR cell is similar. To improve the camping probability of UEs in SA networking, NSA-to-SA switching based on downlink user experience will be initiated for the UE in this situation. [Figure 4-5](#) shows the procedure. This function is enabled by selecting the following options on the eNodeB and source gNodeB for handover:

- **DL_NSA_TO_SA_SWITCH_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter
- **LARGE_PKT_UE_IDENT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch** parameter

- **DL_NSA_TO_SA_SWITCH_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter

In addition to these three options, the **NR_TO_LTE_BIG_PKT_STA_TRANS_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter must also be selected on the target gNodeB for handover to allow the eNodeB to determine whether a UE is a large-packet UE.

Figure 4-5 NSA-to-SA switching based on downlink user experience for a UE performing small-packet downlink services



The procedure is as follows:

1. The eNodeB checks whether the function trigger conditions are met.
When a UE is handed over from the gNodeB to the eNodeB due to SA-to-NSA switching based on downlink traffic volume, the eNodeB starts a protection timer (max (**CellHoParaCfg.NrResOptHolnProtectTmr**, 5s)) to prevent ping-pong handovers and checks whether the following conditions are met.
 - The timer has expired.

- The UE has transitioned from the large-packet state to the small-packet state in the downlink, and stays in the small-packet state in the uplink as well.

If both preceding conditions are met, the eNodeB proceeds to step 2. Otherwise, the procedure ends.

2. The eNodeB delivers inter-RAT B1 measurement configurations.
The eNodeB delivers inter-RAT B1 measurement configurations to the UE and starts a handover waiting timer (fixed at 2s). In these configurations, the frequencies are arranged in descending order of priority.
The eNodeB determines the frequencies to be included in measurement configurations, based on the available bands indicated in the NR capabilities of the UE and the **NrNFreq.SaPccFreqPriority** parameter setting.
3. The eNodeB takes different actions depending on whether inter-RAT B1 measurement reports are received before the handover waiting timer expires.
 - If B1 measurement reports are received on all frequencies before the timer expires, the eNodeB deletes the B1 measurement configurations and proceeds to step 4.
 - If B1 measurement reports are received on some frequencies before the timer expires, the eNodeB deletes the B1 measurement configurations after the timer expires and then proceeds to step 4.
 - If no measurement reports are received on any frequency before the timer expires, the eNodeB deletes the B1 measurement configurations, and the procedure ends.

NOTE

If a B1 measurement report is received on the highest-priority frequency before the timer expires, the eNodeB directly proceeds to step 4 without waiting for measurement reports on other frequencies. It does not delete the B1 measurement configurations when receiving such a report and proceeding to the next step. If the highest-priority frequency is qualified for handover, the eNodeB ends the procedure and performs the handover. If the highest-priority frequency is not qualified for handover, the eNodeB performs step 4 for other frequencies on which measurement reports are received. If no measurement reports are received on other frequencies, the procedure ends.

4. The eNodeB evaluates NR cells based on the received B1 measurement reports.
The eNodeB selects the frequency with the highest SA PCC frequency priority (specified by **NrNFreq.SaPccFreqPriority**) among all NR frequencies on which measurement reports are received, and then selects the cell with the highest RSRP on that frequency as the target NR cell. If multiple cells meet the handover requirements, one of them is randomly selected as the target NR cell.
5. The UE executes the handover.
The eNodeB instructs the UE to initiate a handover to the target NR cell.

4.2.2 Network Analysis

4.2.2.1 Benefits

After NSA-to-SA switching based on downlink user experience takes effect, the average downlink UE throughput (indicated by **User Downlink Average**

Throughput (DU) increases. It is recommended that this function be enabled on all base stations that meet the activation conditions in the target area.

Scenarios or factors that affect the gains of this function:

- Larger NR cell bandwidths and better NR coverage lead to higher increases in the average downlink UE throughput after NSA-to-SA switching.
- When available resources in an NR cell are limited, for example, when a large number of UEs are handed over to the NR cell simultaneously, the improvement in average downlink UE throughput after NSA-to-SA switching is insignificant.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

NR side:

- The downlink RB usage in the NR cell, indicated by **Downlink Resource Block Utilizing Rate (DU)**, increases.
- The downlink traffic volume in the NR cell, indicated by **Downlink Traffic Volume (DU)**, increases.
- During NSA-to-SA switching, the average downlink UE throughput, indicated by **User Downlink Average Throughput (DU)**, temporarily decreases.
- After a UE switches from NSA to SA, its transmit power may increase, which can affect the average uplink UE throughput, indicated by **User Uplink Average Throughput (DU)**, when the UE is located at the cell edge.
- The proportion of UEs camping in SA networking increases.
- The average uplink or downlink UE throughput decreases when NR cell channel quality deteriorates or the number of available RBs decreases, for example, when a large number of UEs are handed over to the NR cell simultaneously.

LTE side:

- The downlink RB usage in the LTE cell, indicated by **Downlink Resource Block Utilizing Rate**, decreases.
- The downlink traffic volume in the LTE cell, indicated by **Downlink Traffic Volume**, decreases.
- The downlink RB usage of NSA DC UEs, indicated by **Downlink Resource Block Utilizing Rate (NSA DC)**, decreases.

4.2.3 Requirements

4.2.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-1712 22	NSA/SA Selection Based on User Experience (LTE FDD)	LT1S0NSAS A00	eNodeB	per eNodeB
LTE TDD	MRFD-1712 32	NSA/SA Selection Based on User Experience (LTE TDD)	LT4SCSEHQ TDD	eNodeB	per eNodeB
NR	MRFD-1712 62	NSA/SA Selection Based on User Experience (NR)	NR0S0NSB UE00	gNodeB	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD LTE FDD LTE TDD	SA-to-NSA switching based on downlink traffic volume	• NSA_SA_DL_SEL_OPT_SW option of the gNodeBParameter.NetworkingOptionOptSw parameter • LTE_FDD_NSA_SA_DL_SEL_OPT_SW or LTE_TDD_NSA_SA_DL_SEL_OPT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

None

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Main control boards: Only UMPTe, UMPTg/ga, and UMPTha series boards support this function.

Baseband processing units: Only UBBPe, UBBPf/fw, UBBPg, and UBBPh series boards support this function.

RF Modules

No requirements

4.2.3.4 Networking

None

4.2.3.5 Others

- UE requirements
 - UEs must comply with 3GPP Release 15 or later and support the following capabilities:
 - Capabilities indicated by *eutra-EPC-HO-ToNR-TDD-FR1-r15* and *sa-NR-r15* if the target cell operates in an FR1 TDD frequency band
 - Capabilities indicated by *eutra-EPC-HO-ToNR-FDD-FR1-r15* and *sa-NR-r15* if the target cell operates in an FR1 FDD frequency band
 - This function applies only to UEs that have not set up specific bearers. (These bearers are non-typical eMBB service bearers, such as voice service bearers. For details about bearers, see *QoS Management*.) Specific bearers correspond to QClis 1, 2, 3, 4, 65, 66, 69, 70, 75, and 79.
- Core network requirements
 - The core network must comply with 3GPP Release 15 or later.
 - The N26 interface must be already configured between the MME and AMF.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

[Table 4-11](#) and [Table 4-12](#) describe the parameters related to NSA-to-SA switching based on downlink user experience.

Table 4-11 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Ext Switch	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	Select the DL_NSA_TO_SA_SWITCH_SW option.
Network Performance Optimization Extend Sw	EnodebAlgoExtSwitch.NetworkPrfmOptExtSwitch	Select the LARGE_PKT_UE_IDENT_SW option.
Downlink NSA To SA Switch A2 Rsrp Threshold	NsaDcMgmtConfig.DlNsaToSaSwitchA2RsrpThld	Set this parameter based on the network plan.
NR Res Opt Handover-in Protect Timer	CellHoParaCfg.NrResOptHolnProtectTmr	Set this parameter based on the network plan.
NR Downlink Bandwidth	NrNFreq.NrDlBandwidth	Set this parameter based on the network plan.
NR Cell Downlink Experience Coefficient	NsaDcMgmtConfig.NrCellDlExperienceCoeff	Set this parameter based on the network plan.
SA PCC Frequency Priority	NrNFreq.SaPccFreqPriority	Set this parameter based on the network plan.
ANR Function Switch	CellAlgoSwitch.AnrFunctionSwitch	It is recommended that the NR_EVENT_ANR_SW option be selected when UEs need to be handed over the neighboring NR cell with the highest frequency priority and highest RSRP but the eNodeB does not have neighbor relationships with that cell.

Table 4-12 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Networking Option Opt Sw	gNodeBParam.NetworkingOptionOptSw	Select the DL_NSA_TO_SA_SWITCH_SW option. You are advised to enable this function in contiguous networking scenarios. Therefore, you are advised to select this option for all gNodeBs in the region.

Parameter Name	Parameter ID	Setting Notes
Networking Option Opt Sw	gNodeBParam.N etworkingOption OptSw	Select the NR_TO_LTE_BIG_PKT_STA_TRANS_SW option. You are advised to enable this function in contiguous networking scenarios. Therefore, you are advised to select this option for all gNodeBs in the region.

4.2.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must be already completed before activating this function. For details, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* > "Using MML Commands" and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* > "Basic Mobility Management Procedure for UEs in Connected Mode" > "Operation and Maintenance (E-UTRAN)" > "Using MML Commands".

Activation Command Examples

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//On the eNodeB side
//Selecting the DL_NSA_TO_SA_SWITCH_SW and LARGE_PKT_UE_IDENT_SW options
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=DL_NSA_TO_SA_SWITCH_SW-1,
NetworkPrfmOptExtSwitch=LARGE_PKT_UE_IDENT_SW-1;
//Setting DLNsaToSaSwitchA2RsrpThld
MOD NSADCMGMTCONFIG: LocalCellId=0, DLNsaToSaSwitchA2RsrpThld=-104;
//Setting NrCellDlExperienceCoeff
MOD NSADCMGMTCONFIG: LocalCellId=0, NrCellDlExperienceCoeff=150;
//Setting NR cell downlink bandwidth and SA PCC frequency priority
MOD NRNFREQ: LocalCellId=0, DlArfcn=151600, NrDlBandwidth=CELL_BW_100M, SaPccFreqPriority=1;
//Setting NrResOptHoInProtectTmr
MOD CELLHOPARACFG: LocalCellId=0, NrResOptHoInProtectTmr=10;
//Optional: Selecting the NR_EVENT_ANR_SW option when UEs need to be handed over to the
//neighboring NR cell with the highest frequency priority and highest RSRP but the eNodeB does not have
//neighbor relationships with that cell
MOD CELLALGOSWITCH: LocalCellId=0, AnrFunctionSwitch=NR_EVENT_ANR_SW-1;

//On the gNodeB side
//Selecting the DL_NSA_TO_SA_SWITCH_SW and NR_TO_LTE_BIG_PKT_STA_TRANS_SW options
MOD GNODEBPARAM:
NetworkingOptionOptSw=DL_NSA_TO_SA_SWITCH_SW-1&NR_TO_LTE_BIG_PKT_STA_TRANS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the eNodeB side
//(Optional) Deselecting the NR_EVENT_ANR_SW option
MOD CELLALGOSWITCH: LocalCellId=0, AnrFunctionSwitch=NR_EVENT_ANR_SW-0;
//Deselecting the DL_NSA_TO_SA_SWITCH_SW and LARGE_PKT_UE_IDENT_SW options
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=DL_NSA_TO_SA_SWITCH_SW-0,
NetworkPrfmOptExtSwitch=LARGE_PKT_UE_IDENT_SW-0;

//On the gNodeB side
//Deselecting the DL_NSA_TO_SA_SWITCH_SW and NR_TO_LTE_BIG_PKT_STA_TRANS_SW options
MOD GNODEBPARAM:
NetworkingOptionOptSw=DL_NSA_TO_SA_SWITCH_SW-0&NR_TO_LTE_BIG_PKT_STA_TRANS_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

If any of the counters listed in [Table 4-13](#) is not 0, NSA/SA selection based on downlink user experience has taken effect.

Table 4-13 Counters related to NSA-to-SA switching based on downlink user experience

Counter ID	Counter Name
1526783875	L.IRATHO.E2N.DIExpOptSel.PrepAttOut
1526783876	L.IRATHO.E2N.DIExpOptSel.ExecAttOut
1526783877	L.IRATHO.E2N.DIExpOptSel.ExecSuccOut

4.2.4.3 Network Monitoring

To evaluate the performance of NSA-to-SA switching based on downlink user experience, monitor the counters listed in [Table 4-14](#).

Table 4-14 Counters used to monitor NSA-to-SA switching based on downlink user experience

Counter ID	Counter Name
1526783875	L.IRATHO.E2N.DIExpOptSel.PrepAttOut
1526783876	L.IRATHO.E2N.DIExpOptSel.ExecAttOut

Counter ID	Counter Name
1526783877	L.IRATHO.E2N.DIExpOptSel.ExecSuccOut
1911820677	N.HO.InterRAT.E2N.PrepAttIn
1911820678	N.HO.InterRAT.E2N.PrepSuccIn
1911820679	N.HO.InterRAT.E2N.ExecSuccIn

5 NSA/SA Selection Based on Uplink Coverage

5.1 Principles

NSA/SA selection based on uplink coverage enables the **gNodeB to identify UEs under weak uplink coverage and trigger SA-to-NSA switching for UEs based on user experience. This function provides a networking scheme with better service experience for UEs, thus achieving the goal of optimal user experience. This function works only in scenarios with NR TDD cells and LTE FDD cells.**

This function includes the following sub-functions:

- SA-to-NSA switching based on weak uplink coverage
- NSA-to-SA switching based on strong uplink coverage

NSA/SA selection based on uplink coverage is enabled when all of the following conditions are met:

- On the gNodeB side:
 - The **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch.InterRatServiceMobilitySw** parameter is selected to enable inter-RAT mobility from NG-RAN to E-UTRAN.
 - The **NSA_SA_UL_SEL_OPT_SW** options of the **gNBOperator.OperatorInterRatPolicySw** and **gNodeBParam.NetworkingOptionOptSw** parameters are selected to enable NSA/SA selection based on uplink user experience.
The **NSA_SA_UL_SEL_OPT_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter takes effect only if the **NSA_SA_UL_SEL_OPT_SW** option of the **gNBOperator.OperatorInterRatPolicySw** parameter is selected.
 - The **EUTRAN_HO_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter is selected.
 - The **NSA_SA_UL_WEAK_FRBD_SW** option of the **NRInterRatHoParam.MobilityHOControlSwitch** parameter is deselected

for the target cell. If this option is selected for the target cell, SA-to-NSA switching based on weak uplink coverage will not take effect.

- (Optional) In the RAN sharing with common carrier scenario (where the **gNBSharingMode.gNBMultiOpSharingMode** parameter is set to **SHARED_FREQ**), inter-RAT mobility policies are configured based on the operator by selecting the **EUTRAN_HO_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter.
- (Optional) In inter-gNodeB CA scenarios, the **NSA_SA_SEL_INTER_GNB_CA_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter is selected, so as to enable NSA/SA selection based on user experience for inter-gNodeB CA UEs.
- On the eNodeB side:
The **LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter is selected.

NSA/SA selection based on uplink coverage requires a UE to report its NR, EUTRA-NR, and EUTRA capabilities through a UE Capability Information message. If all these capabilities are included, the message length may exceed the maximum defined in 3GPP specifications. As a result, this message may be truncated, which can affect NSA/SA selection based on uplink coverage. When the UE capability separate query function is enabled (by selecting the **UE_CAP_SEPARATE_QUERY_SW** option of the **NRCellAlgoSwitch.ProcessSwitch** parameter), the gNodeB can query NR, EUTRA-NR, and EUTRA capabilities separately to ensure that the UE's complete capability information is obtained. This function supports blacklist control. When the **UE_CAPB_SEPARATE_QUERY_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlSwitch** parameter is selected, the preceding function does not take effect for blacklisted UEs.

5.1.1 SA-to-NSA Switching Based on Weak Uplink Coverage

The gNodeB checks the uplink coverage of UEs. For UEs under weak uplink coverage, the gNodeB initiates SA-to-NSA switching.

This function includes two steps:

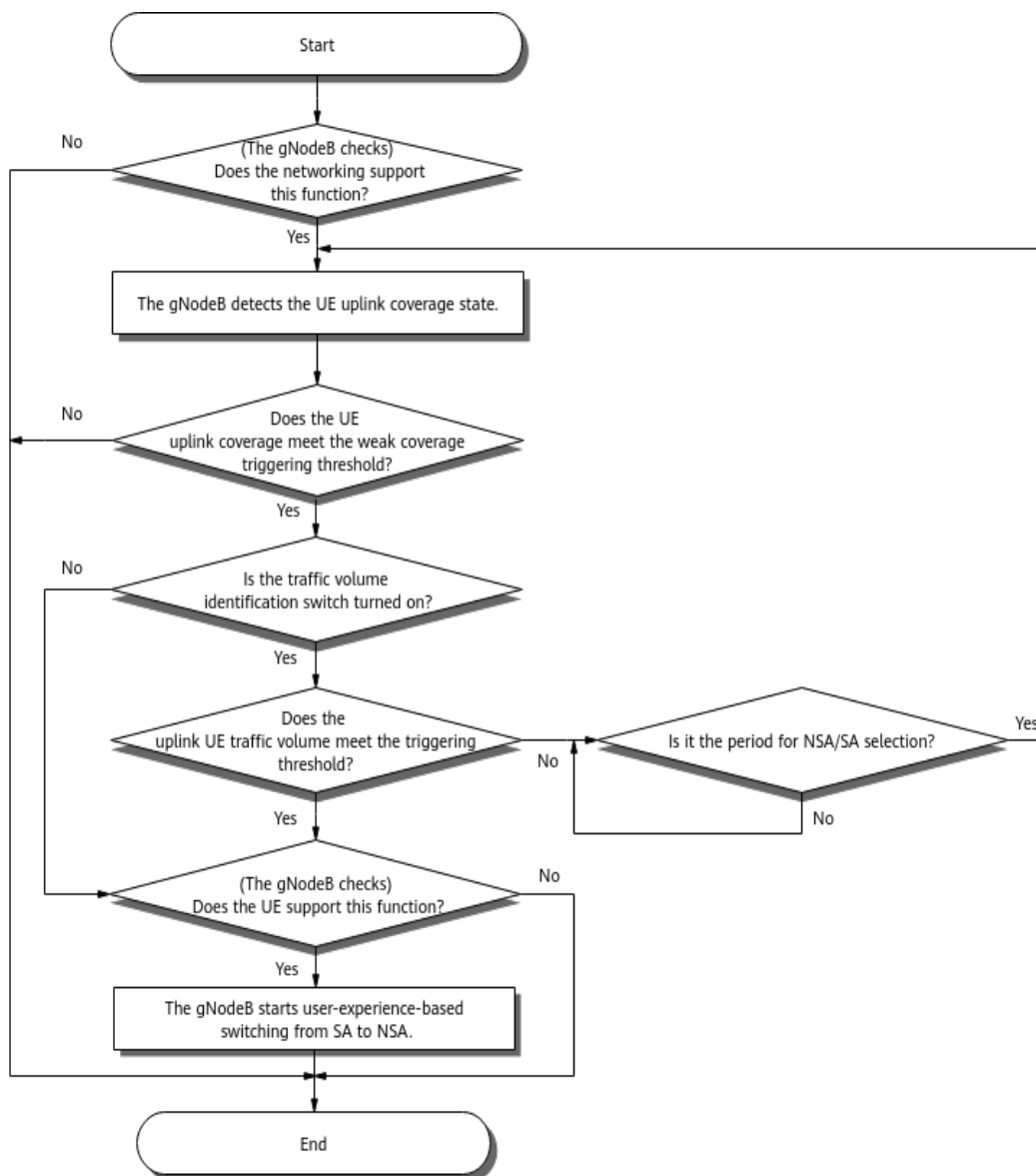
1. The gNodeB identifies UEs under weak uplink coverage. For details, see [5.1.1.1 Identifying UEs Under Weak Uplink Coverage](#).
2. The gNodeB initiates SA-to-NSA switching based on user experience for the UEs under weak uplink coverage. Currently, the following policies are supported based on different RSRP obtaining methods:
 - **SA-to-NSA switching based on UE measurements:** The gNodeB decides and executes a handover based on the RSRP from UE measurements. For details, see [5.1.1.2 SA-to-NSA Switching Based on UE Measurements](#).
 - **SA-to-NSA switching with RSRP predicted based on virtual grids:** The RSRP predicted using the NL RSRP model is used instead of the RSRP from UE measurements for handover decision and execution. For details, see [5.1.1.3 SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids](#).

After a UE accesses an NSA network, **E-UTRAN to NG-RAN traffic steering or fast return from E-UTRAN to NG-RAN** allows the UE to quickly return to an SA network. For details, see [5.1.1.4 Handling After UE Access in NSA](#).

5.1.1.1 Identifying UEs Under Weak Uplink Coverage

The procedure for the gNodeB to identify UEs under weak uplink coverage is shown in [Figure 5-1](#).

Figure 5-1 Procedure for identifying UEs under weak uplink coverage



The procedure is as follows:

1. The gNodeB checks whether the networking supports this function
The NR cell must operate in NSA and SA hybrid networking. Otherwise, none of the UEs in the NR cell supports this function, and the procedure ends.

2. The gNodeB checks the uplink coverage of a UE and determines whether the UE is under weak coverage. The coverage status of all UEs is initialized to **under strong coverage**.

The gNodeB checks the real-time uplink coverage of all UEs in the cell.

- If the gNodeB detects that the uplink SRS SINR of a UE is less than the **NRDUCellSrsMeas.NrToEutranSinnrLowThld** parameter value minus 2 dB for five consecutive times, the UE is considered to be under weak coverage. The detection interval is specified by the **NRDUCellSrsMeas.NsaULPathSelSinnrTimeToTrig** parameter.
- Otherwise, the UE remains under strong coverage, and the procedure ends.

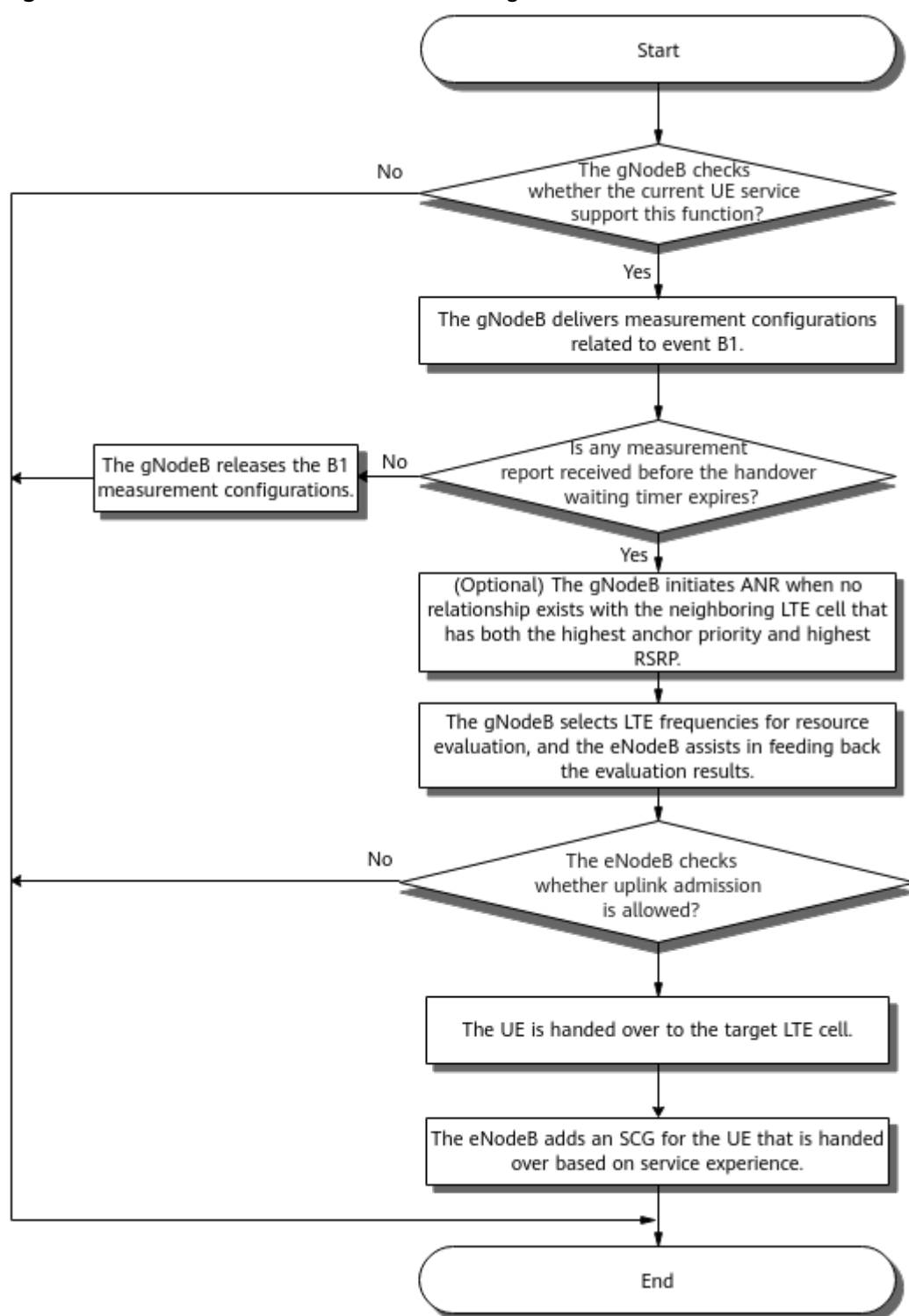
Real-time uplink coverage detection is not performed for UEs with uplink CA configured or UEs supporting super uplink. Therefore, these UEs remain under strong coverage, and the procedure ends.

3. If the **VOLUME_IDENTIFY_SW** option of the **NRCCellAlgoSwitch.ServiceFunctionSwitch** parameter is selected, the procedure goes to the next step when the UE is under weak coverage. Otherwise, the procedure goes to step 5.
4. The gNodeB checks the accumulated volume of to-be-transmitted data of the UE:
 - If the accumulated data volume is greater than the **NRDUCellServExp.CommonULLargePktIdentThld** parameter value, the UE is identified as a common large-packet UE in the uplink.
 - Otherwise, the UE is not identified as a common large-packet UE in the uplink. For such UEs, when the interval specified by the **gNodeBParam.NsaSaSelOptPeriod** parameter arrives, the procedure for identifying UEs under weak uplink coverage continues, that is, step 2 is performed.
5. The gNodeB checks whether the UE complies with 3GPP Release 15 or later, as well as whether the UE supports SA, and handoverLTE-EPC and EN-DC capabilities. If yes, the procedure goes to the next step. If no, the procedure ends.
6. The gNodeB starts user-experience-based switching from SA to NSA.

5.1.1.2 SA-to-NSA Switching Based on UE Measurements

The gNodeB starts SA-to-NSA switching based on UE measurements for UEs under weak uplink coverage, as shown in [Figure 5-2](#).

Figure 5-2 Procedure for SA-to-NSA switching based on UE measurements



The procedure is as follows:

1. The gNodeB checks the network option optimization policies for the QCI of services running on a UE.

The **gNBQciBearer.NetworkingOptionOptFlag** parameter specifies the policy. The following values are possible:

- **FORBID**: Network option optimization is not performed for a UE running services with the specified QCI.
- **PERMIT**: Network option optimization is allowed for a UE running services with the specified QCI. However, if services with a QCI for which this parameter is set to **FORBID** are also running on the UE, network option optimization is not performed for the UE.
- **FOLLOW**: Network option optimization is not performed for a UE running only the services with the specified QCIs. However, if services with other QCIs for which this parameter is set to a value other than **FOLLOW** are also running on the UE, network option optimization complies with the policies defined for such QCIs.

According to the Networking Option Opt Flag settings corresponding to the QCIs of services running on the UE, the gNodeB takes the following actions:

- If a UE is running a service with a QCI for which the parameter is set to **PERMIT** and with no QCIs of services set to **FORBID**, the procedure goes to the next step.
- If a UE is running a service with a QCI for which the parameter is set to **FORBID** or running only services with QCIs for which the parameter is set to **FOLLOW**, the procedure ends, that is, SA-to-NSA switching based on weak uplink coverage does not take effect.

When flexible user steering is enabled (that is, when the **gNB RfSP Config.NetworkingOptionOptFlag** parameter is set to **TRUE** to allow customized network option optimization policies for UEs based on the RFSP), and RFSP-based NG-RAN-to-E-UTRAN handover is enabled (by selecting the **NR_TO_EUTRAN_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter), the customized network option optimization policies take effect.

2. The gNodeB delivers measurement configurations related to inter-RAT event B1.

The gNodeB includes the frequencies in the descending order of priority in the measurement configurations related to inter-RAT event B1 (namely that for network architecture selection), delivers them to the UE, and starts a handover waiting timer, whose length is always 2s. When the measured RSRP value of an E-UTRAN cell exceeds the *Thresh* (specified by the **NRCellHoEutranMeaGrp.NetworkingOptionOptB1Thld** parameter), the measurement report of event B1 will be sent.

The gNodeB determines the frequencies to be measured and their priorities based on the following principles:

- The gNodeB determines the frequencies to be included in measurement configurations, based on the available band combinations indicated in the EN-DC capabilities of the UE and the **NSA PCC Frequency Priority** specified by the **NRCellEutranNFreq.NsaPccFreqPriority** parameter.

A frequency that meets one of the following conditions cannot be selected as a candidate:

- The **NRCellEutranNFreq.FreqIntrfFlag** parameter is set to **INTRF** for the frequency.
- The **NRCellEutranNFreq.FreqRelationship** parameter is set to **OVERLAPPED** for the frequency, and the

NSA_FREQ_CONFLICT_AVOID_SW option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.

- The gNodeB can deliver a maximum of six frequencies for measurement. The frequencies that can be selected as PCC frequencies are delivered preferentially. If the number of such frequencies does not reach six, the gNodeB delivers frequencies that can be selected as SCC frequencies as a supplement. The gNodeB arranges the PCC frequencies in descending order of their NSA PCC frequency priorities and arranges the SCC frequencies in descending order of the NSA PCC frequency priorities for the corresponding PCC frequencies. If multiple frequencies have the same priority, the frequencies with the smaller absolute radio frequency channel numbers are preferentially selected.
3. The gNodeB takes different actions, depending on whether it has received inter-RAT B1 measurement reports before the handover waiting timer expires.
- If the gNodeB receives inter-RAT B1 measurement reports of all frequencies before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations immediately and goes to the next step.
 - If the gNodeB receives inter-RAT B1 measurement reports of some of the frequencies before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations after the expiry and goes to the next step.
 - If the gNodeB does not receive measurement reports of any frequency before the handover waiting timer expires, the gNodeB releases the B1 measurement configurations. The procedure ends.
4. (Optional) To support handovers from SA to the UE-supported but missed neighboring LTE cell with the highest anchor priority (specified by **NRCeUtranNFreq.NsaPccFreqPriority**) and highest RSRP after the gNodeB has received all LTE B1 measurement reports, the **NSA_SA_SEL_MISS_CELL_ANR_SW** option of the **NRCeUtranAnr.AnrAlgoSwitch** parameter can be selected to enable delivery of CGI measurement configurations. The gNodeB then determines whether to add missed neighbor relationships based on the CGI measurement results according to the configured ANR mode. Specifically:
- In ANR controlled mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCeUtranAnr.AnrModeSwitch** parameter selected):
 - If handovers are allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCeUtranAnr.AnrModeSwitch** parameter selected): CGI measurement configuration can be delivered. The neighboring LTE cell is added as a to-be-confirmed neighboring cell based on the CGI measurement result, and SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained in 6.
 - If handovers are not allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCeUtranAnr.AnrModeSwitch** parameter deselected): Handovers to the to-be-confirmed neighboring cell is not allowed, but CGI measurement configurations are still delivered once. After this neighboring cell is confirmed as a formal one on the OSS, handovers to this neighboring cell can be performed.

- In ANR free mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCCellAnr.AnrModeSwitch** parameter deselected): CGI measurement configurations can be delivered. After receiving a CGI measurement report, the gNodeB adds the neighboring cell as a formal one based on the CGI measurement result. SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained in 6.

If the **NSA_SA_SEL_MISS_CELL_ANR_SW** option is deselected, step 5 is performed after step 3 is complete.

5. The gNodeB selects LTE frequencies for resource evaluation, and the eNodeB assists in feeding back the evaluation results.

The gNodeB preferentially selects the eNodeB serving the cell with the highest RSRP value on the frequency with the highest NSA PCC frequency priority. Then, the gNodeB checks whether an X2 interface is available between itself and this eNodeB.

- If an X2 interface is available between the gNodeB and this eNodeB, the gNodeB sends an EN-DC Private Evaluation Request message, which is an uplink admission request message, over the X2 interface to request the result of uplink admission to the PCC, and proceeds to the next step.
- If an X2 interface is unavailable between the gNodeB and this eNodeB, the gNodeB selects the eNodeB serving the cell with the highest RSRP value on the frequency with the second highest NSA PCC frequency priority. Then, the gNodeB checks whether an X2 interface is available between itself and this eNodeB. The gNodeB repeats the selection activities until an appropriate eNodeB is found. If no qualified eNodeB is found, the procedure ends.

If there are multiple frequencies with the same NSA PCC frequency priority, the eNodeB serving the cell with the highest RSRP value on the frequency with the smallest absolute radio frequency channel number is selected for the following steps.

6. The gNodeB sends an uplink admission request message to the eNodeB (for checking whether this cell can be treated a PCell). After receiving the message, the eNodeB evaluates whether to accept the admission request based on the uplink PRB usage, number of active UEs, and uplink interference in the LTE cell. A cell that meets any of the following conditions cannot be selected as a PCell:

- Uplink PRB usage > **NsaDcMgmtConfig.SaToNsaPccUlPrbUsageThld**
- Number of active UEs > **NsaDcMgmtConfig.SaToNsaPccActiveUeNumThld**
- Uplink interference > **NsaDcMgmtConfig.SaToNsaPccUlIntrfThld**

If multiple cells can act as a PCell, the eNodeB selects the first qualified cell as the PCell. The gNodeB checks the uplink admission results returned by the eNodeB and makes a decision:

- If the eNodeB accepts the admission request, it responds with the information about the LTE cell. The procedure goes to the next step.
- If the eNodeB rejects the admission request, the procedure ends.

7. The UE is handed over to the target LTE cell.

The gNodeB initiates a handover to the target LTE cell and sends the information about the target LTE cell and the serving NR cell to the eNodeB

through a handover request. The serving NR cell is used as the PSCell to be added for the target LTE cell.

Then the UE is handed over to the target LTE cell. In this case, the incoming handover is caused by SA-to-NSA switching based on user experience, and the following operations are performed:

- If the UE supports direct SA-to-NSA switching and the **DIRECT_NR_HO_TO_EN_DC_SW** option of the eNodeB parameter **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** is selected, the eNodeB directly adds an SCG based on the LTE and NR cell information included in the handover request, enabling the UE to switch from the NR to EN-DC state. For details, see *EPC-based NSA Performance Enhancement*.
- Otherwise, an SCG can be added to set up EN-DC services only after the UE is handed over to the LTE cell.

 NOTE

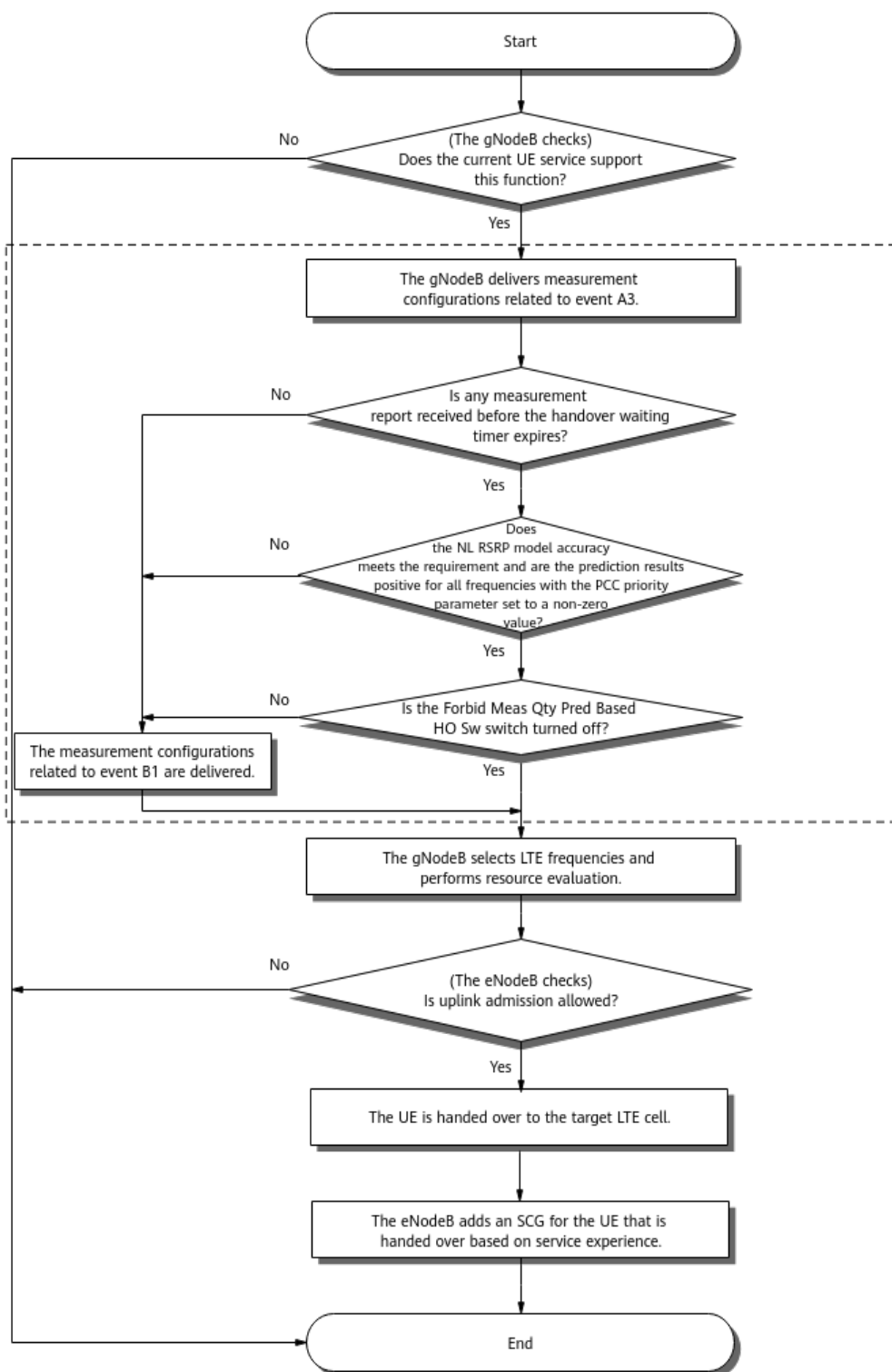
When a UE accesses an NSA network with user-experience-based switching from SA to NSA, uplink data is transmitted only on the master cell group (MCG) side.

5.1.1.3 SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids

During NSA/SA selection based on weak uplink coverage, the RSRP predicted by the NL RSRP model can be used to replace the RSRP measured by the UE for handover decision making. This function is controlled by the **NSA_SA_SEL_OPT_RSRP_PRED_SW** option of the **NRCelAlgoSwitch.InterRatServiceMobilitySw** parameter. (For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.)

Figure 5-3 shows the procedure for SA-to-NSA switching with RSRP predicted based on virtual grids.

Figure 5-3 Procedure for SA-to-NSA switching with RSRP predicted based on virtual grids



Note: The content in the dashed-line box in the figure is the difference between the NSA/SA selection with RSRP predicted based on virtual grids and that based on UE measurements.

This function differs from SA-to-NSA switching based on UE measurements, as shown in [Figure 5-2](#), only in the delivery of event measurement configurations by the gNodeB, the obtaining of RSRP, and timer processing:

1. During SA-to-NSA switching with RSRP predicted based on virtual grids, the gNodeB delivers **A3 measurement configurations** related to the frequency on which a UE camps. If the UE reports an A3 measurement report within 1s (that is, before the handover waiting timer expires), the gNodeB uses the reported measurement result (that is, the RSRP values of the serving cell and intra-frequency neighboring cells) to determine the coverage status of candidate LTE frequencies.
2. The RSRP for a neighboring frequency can be predicted using the NL RSRP model, without the need for UE measurements, when all the following conditions are met:
 - The accuracy of the NL RSRP model on the gNodeB meets the accuracy requirement. Whether the model accuracy meets the requirement is determined as follows:
 - When the model accuracy is greater than or equal to the **NRCellSmartMultiCarr.PredAccuForDataThld** parameter value, the accuracy requirement is met and the grid model can be used for data service mobility procedures.
 - When the model accuracy is lower than the **NRCellSmartMultiCarr.PredAccuForDataThld** parameter value, the accuracy requirement is not met, the grid model cannot be used for data service mobility procedures, and the measurement control procedure related to event B1 is performed, as shown in [3](#).
 - For all LTE frequencies with the **NRCellEutranNFreq.NsaPccFreqPriority** parameter set to non-zero values, the results of prediction of using the NL RSRP model are either positive or negative.
 - The **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr.HoProtectAlgoSwitch** parameter is deselected.
3. During SA-to-NSA switching with RSRP predicted based on virtual grids, the gNodeB delivers **B1 measurement configurations** to UEs if any of the following conditions is met:
 - The UE does not send an A3 measurement report within 1s.
 - The **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr.HoProtectAlgoSwitch** parameter is selected.
 - The model accuracy does not meet the accuracy requirement of the algorithm, that is, the model accuracy is lower than the **NRCellSmartMultiCarr.PredAccuForDataThld** parameter value.
 - Among all LTE frequencies with the **NRCellEutranNFreq.NsaPccFreqPriority** parameter set to non-zero values, the results of prediction using the NL RSRP model are not clear, that is, neither positive nor negative, for at least one of these frequencies.

5.1.1.4 Handling After UE Access in NSA

After a UE accesses an NSA network, E-UTRAN to NG-RAN traffic steering or fast return from E-UTRAN to NG-RAN allows the UE to quickly return to an SA network.

- If flexible user steering is not enabled, the traffic steering or fast return procedure back to SA can be performed when the **ServiceIrHoCfgGroup.InterRatHoState** parameter is set to **MUST_HO** for at least one QCI among the QCIs of services running on the UE.
- If flexible user steering is also enabled:
 - When the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidAlgoSwitch** parameter is deselected for the UE, the traffic steering or fast return procedure can be performed if no **Inter-RAT handover state** is set to **NO_HO** for all of the QCIs of services running on the UE.
 - When the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidAlgoSwitch** parameter is selected for the UE, the traffic steering or fast return procedure can be performed if the **Inter-RAT handover state** is set to **MUST_HO** for at least one QCI among the QCIs of services running on the UE.

If the UE is not expected to return to an SA network for anti-ping-pong consideration, the following parameter setting policies are recommended:

- If flexible user steering is not enabled, **Inter-RAT handover state** cannot be set to **MUST_HO** for the QCIs of services running on the UE.
- If flexible user steering is also enabled:
 - If the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidAlgoSwitch** parameter is deselected for the UE, the **Inter-RAT handover state** must be set to **NO_HO** for at least one QCI among the QCIs of services running on the UE.
 - If the **NSA_SA_SEL_OPT_SW** option of the **SpidCfg.SpidAlgoSwitch** parameter is selected for the UE, the **Inter-RAT handover state** cannot be set to **MUST_HO** for any QCI of services running on the UE.

5.1.2 NSA-to-SA Switching Based on Strong Uplink Coverage

The gNodeB checks the uplink coverage of UEs. For UEs under strong uplink coverage, the gNodeB initiates NSA-to-SA switching.

This function includes two steps:

1. The gNodeB identifies UEs under strong uplink coverage. For details, see [5.1.2.1 Identifying UEs Under Strong Uplink Coverage](#).
2. The gNodeB initiates NSA-to-SA switching based on strong uplink coverage for the UEs. For details, see [5.1.2.2 NSA-to-SA Switching Based on UE Measurements](#).

5.1.2.1 Identifying UEs Under Strong Uplink Coverage

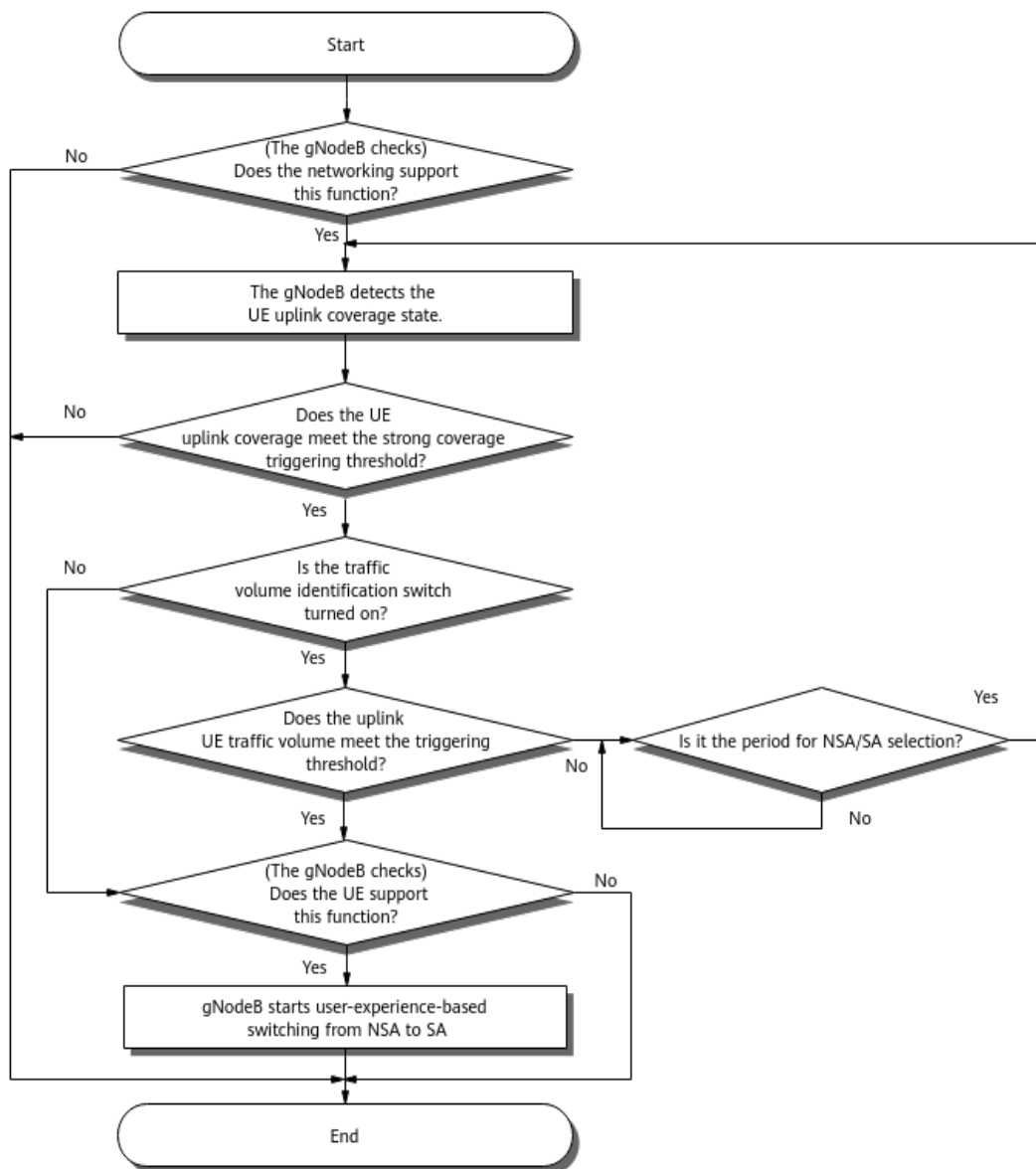
The overall procedure for identifying UEs under strong uplink coverage, as shown in [Figure 5-4](#), is similar to that for identifying UEs under weak uplink coverage, as shown in [Figure 5-1](#). The differences are as follows:

1. The gNodeB checks the uplink coverage of a UE and determines whether the UE is under strong coverage.

The gNodeB checks the real-time uplink coverage of all UEs in the cell. The coverage status of all UEs is initialized to **weak coverage**.

- If the gNodeB detects that the uplink SRS SINR of a UE is greater than the **NRDUCellSrsMeas.EutranToNrSinnrHighThld** parameter value plus 2 dB for five consecutive times, the UE is considered to be under strong coverage. The detection interval is specified by the **NRDUCellSrsMeas.NsaUlPathSelSinnrTimeToTrig** parameter.
 - Otherwise, the UE remains to be recorded as under weak coverage.
2. The gNodeB starts the NSA-to-SA switching based on user experience.

Figure 5-4 Procedure for identifying UEs under strong uplink coverage



5.1.2.2 NSA-to-SA Switching Based on UE Measurements

For a UE under strong coverage, the procedure for NSA-to-SA switching based on UE measurements is as follows:

1. The gNodeB delivers A1 measurement configurations to check the signal quality of the serving NR cell.

- If the gNodeB receives A1 measurement reports, the signal quality of the NR cell is satisfactory. The gNodeB sends an X2 message to instruct the eNodeB to perform a handover for NSA-to-SA switching.
- If the gNodeB does not receive A1 measurement reports, the signal quality of the NR cell is unsatisfactory. The procedure ends.

NOTE

Threshold for reporting event A1 = **NRCeInterRHoMeaGrp.InterRatHoA1RsrpThld** + **gNBOperatorQciParam.EutranA1RsrpThldOffset** + **gNBRfspConfig.InterRatA1A2B2RsrpThldOfs** + **NRCeOpPolicy.EutranA1A2B2RsrpThldOfs**

2. After receiving the NSA-to-SA switching request message, the eNodeB performs a handover for the switching.

5.2 Network Analysis

5.2.1 Benefits

After NSA/SA selection based on uplink coverage takes effect, the average uplink UE throughput, indicated by the **User Uplink Average Throughput (DU)**, increases, and the service drop rate in NR cell, indicated by the **Service Call Drop Rate (CU, Inactive)**, decreases.

To obtain greater gains, it is recommended that the **VOLUME_IDENTIFY_SW** option of the **NRCeAlgoSwitch.ServiceFunctionSwitch** parameter be selected as well.

Scenarios or factors that affect the gains of this function:

- The larger the LTE FDD cell bandwidth and the better the coverage, the higher the average uplink UE throughput after SA-to-NSA switching.
- In scenarios where available resources in LTE FDD cells are limited, for example, when handovers of a large number of UEs are initiated towards an LTE cell at the same time, the improvement in the average uplink UE throughput after SA-to-NSA switching is insignificant.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

- SA-to-NSA switching based on weak uplink coverage:
NR side:
 - The downlink RB usage of NR cell, indicated by the **Downlink Resource Block Utilizing Rate (DU)**, decreases.
 - The downlink traffic volume in NR cell, indicated by the **Downlink Traffic Volume (DU)**, decreases.
 - During switching from SA to NSA, for a period of time, the average uplink UE throughput, indicated by the **User Uplink Average Throughput (DU)**, decreases.

- The proportion of UEs camping in the SA network decreases.
- If the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled (by selecting the **UL_TFC_NR_LTE_MOBILITY_ENH_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter), the **N.NsaDc.SgNB.Mod.Req.Att** counter value decreases.
- **Table 5-1** describes the change trends of the number of SA-to-NSA switching times when specific functions are enabled. **Table 5-2** lists the counters related to the number of SA-to-NSA switching times.

Table 5-1 Change trends of the number of SA-to-NSA switching times when specific functions are enabled

Function Name	Change Trend of the Number of SA-to-NSA Switching Times
NSA/SA selection based on user experience for inter-gNodeB CA UEs	Increase
ANR for missing neighboring LTE cells	Increase

Table 5-2 Counters related to the number of SA-to-NSA switching times

Counter ID	Counter Name
1911829365	N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
1911829366	N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
1911829367	N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
1911853804	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.UL
1911853805	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.UL
1911853806	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.UL
1526765179	L.IRATHO.N2E.Direct2EnDc.ExecSuccIn
1526765175	L.IRATHO.N2E.Direct2EnDc.PrepAttIn
1526765176	L.IRATHO.N2E.Direct2EnDc.PrepInSucc
1526749536	L.IRATHO.N2E.ExecSuccIn
1526749534	L.IRATHO.N2E.PrepAttIn
1526749535	L.IRATHO.N2E.PrepInSucc

LTE side:

- If downlink data split is enabled, the uplink RB usage, indicated by the **Uplink Resource Block Utilizing Rate**, and downlink RB usage, indicated by the **Downlink Resource Block Utilizing Rate**, of LTE cells increase.
- If downlink data split is enabled, the uplink traffic volume and downlink traffic volume of LTE cells, indicated by **Uplink Traffic Volume** and **Downlink Traffic Volume**, respectively, increase.
- The downlink RB usage of NSA DC UEs, indicated by the **Downlink Resource Block Utilizing Rate (NSA DC)**, increases.
- If the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled (by selecting the), the **L.NsaDc.SCG.Mod.Req.Att** counter value decreases.
- The average uplink UE throughput decreases if there is uplink interference to LTE cells or the number of available RBs decreases. An example of such scenarios is that the uplink load gets heavier when a large number of UEs are handed over to an LTE cell at the same time.
- NSA-to-SA Switching based on strong uplink coverage:
 - NR side:
 - The downlink RB usage of NR cell, indicated by the **Downlink Resource Block Utilizing Rate (DU)**, increases.
 - The downlink traffic volume in NR cell, indicated by the **Downlink Traffic Volume (DU)**, increases.
 - During switching from NSA to SA, for a period of time, the average uplink UE throughput, indicated by the **User Uplink Average Throughput (DU)** counter, decreases.
 - This function increases the proportion of UEs camping in the SA network.
 - If the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled (by selecting the), the **N.NsaDc.SgNB.Mod.Req.Att** counter value decreases.
 - The average uplink UE throughput decreases if there is uplink interference to an NR cell or the number of available RBs decreases. An example of such scenarios is that the uplink load gets heavier when a large number of UEs are handed over to an NR cell at the same time.
 - After NSA/SA selection based on user experience for inter-gNodeB CA UEs takes effect, the number of SA-to-NSA switching times increases.
 - LTE side:
 - If downlink data split is enabled, the uplink RB usage, indicated by **Uplink Resource Block Utilizing Rate**, and downlink RB usage, indicated by **Downlink Resource Block Utilizing Rate**, of LTE cells decrease.
 - If downlink data split is enabled, the uplink traffic volume and downlink traffic volume of LTE cells, indicated by the **Uplink Traffic Volume** and **Downlink Traffic Volume**, respectively, decrease.
 - The downlink RB usage of NSA DC UEs, indicated by the **Downlink Resource Block Utilizing Rate (NSA DC)**, decreases.
 - If the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled (by selecting the), the **L.NsaDc.SCG.Mod.Req.Att** counter value decreases.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	MRFD-17 1222	NSA/SA Selection Based on User Experience (LTE FDD)	LT1S0NSA SA00	NSA/SA Selection Based on User Experience (LTE FDD)	eNodeB	per eNodeB
NR	MRFD-17 1262	NSA/SA Selection Based on User Experience (NR)	NR0S0NS BUE00	NSA/SA Selection Based on User Experience (NR)	gNodeB	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
Low-frequency NR TDD	NSA networking based on EPC	NRCeIIAlgoSwitch.NsaDcSwitch	<i>EPC-based NSA Basics</i>

RAT	Function Name	Function Switch	Reference
LTE FDD	NSA networking based on EPC	NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgorithm parameter	<i>EPC-based NSA Basics</i>
Low-frequency NR TDD	NSA carrier combination selection based on user experience ^a	NSA_LTE_SEL_OPT_SW option of the gNodeBParam.NsaNetworkingOptionOptSw parameter	<i>EPC-based NSA Experience Improvement</i>
Low-frequency NR TDD	NSA/SA selection based on downlink traffic volume ^a	NSA_SA_DL_SEL_OPT_SW option of the gNodeBParam.NsaNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
Low-frequency NR TDD	NSA/SA selection based on uplink coverage ^a	NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NsaNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
NR FDD Low-frequency NR TDD	NL RSRP model building ^b	LNR_HO_MEAS_QTY_PRED_FUNCTION_SW option of the NRCellSmartMultiCarr.LnrMultiCarrierAlgorithm parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>
NR FDD Low-frequency NR TDD	NSA/SA selection based on downlink traffic volume ^b	NSA_SA_DL_SEL_OPT_SW option of the gNodeBParam.NsaNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>

RAT	Function Name	Function Switch	Reference
Low-frequency NR TDD	NSA-SA selection based on uplink coverage ^b	NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
Low-frequency NR TDD	NSA-SA uplink selection ^b	NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination ^c	VG_MODEL_ALL_OW_BUILD_FLAG option of the NRCCellEutranNFreq.AggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i>
NR FDD Low-frequency NR TDD	Inter-RAT ANR ^d	NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>
Note: a: If the NSA_SA_SEL_INTER_GNB_CA_SW option of the gNodeBParam.NetworkingOptionOptSw parameter is selected, the NSA_LTE_SEL_OPT_SW, NSA_SA_DL_SEL_OPT_SW, or NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter must be selected. b: The NSA_SA_UL_SEL_OPT_SW option of the gNBOperator.OperatorInterRatPolicySw parameter takes effect only if the NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter is selected. c: SA-to-NSA Switching with RSRP Predicted Based on Virtual Grids requires that (1) NL RSRP model building be enabled, (2) either NSA-SA selection based on downlink traffic volume or NSA-SA selection based on uplink coverage be enabled, and (3) Virtual Grid-based Multi-Frequency Coordination be enabled for at least one neighboring LTE frequency of the NR cell. d: The NSA_SA_SEL_MISS_CELL_ANR_SW option can be selected only when the NR_EUTRAN_ANR_SW option is selected.			

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink CA	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	The gNodeB carries the selected EN-DC combination in the selectedbandCombinationInfoEN-DC-v1540 IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA networking. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on uplink coverage is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	E-UTRAN to NG-RAN traffic steering	ServBasedNrHoSwitch option of the CellAlgoSwitch.HoAllowedSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	After NSA/SA selection based on uplink coverage is enabled, the evaluation of E-UTRAN to NG-RAN traffic steering and that of fast return from E-UTRAN to NG-RAN are changed. This results in a lower probability that E-UTRAN to NG-RAN traffic steering and fast return from E-UTRAN to NG-RAN take effect, and therefore smaller values of the L.IRATHO.E2N.Service.PrepareAttOut and L.IRATHO.E2N.FastReturn.PrepareAttOut counters.
LTE FDD	Fast return from E-UTRAN to NG-RAN	FAST_RETURN_TO_NR_SW option of the CellAlgoExtSwitch.HoAllowedSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	
Low-frequency NR TDD	User-experience-based NSA SCG coverage expansion	COV_THLD_ADAPT_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When NSA/SA selection based on uplink coverage is enabled together with user-experience-based NSA SCG coverage expansion, the probability that NSA/SA selection based on uplink coverage takes effect decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	User-experience-based coverage expansion	COV_THL D_ADAPT_SW option of the gNB MobilityComParam. MobilityAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When NSA/SA selection based on uplink coverage is enabled together with user-experience-based coverage expansion, the probability that NSA/SA selection based on uplink coverage takes effect decreases.
Low-frequency NR TDD	Protection of UEs under weak uplink coverage	UL_WEAK_COV_U E_PROTECT_SW option of the NRCellAlgoSwitch. MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When protection of UEs under weak uplink coverage is enabled together with SA-to-NSA switching based on weak uplink coverage and the conditions for SA-to-NSA switching based on weak uplink coverage are met, SA-to-NSA switching based on weak uplink coverage takes effect preferentially. If the conditions for SA-to-NSA handover are not met, protection of UEs under weak uplink coverage takes effect.
LTE FDD	FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs	NsaDcAlgoParam. NrTddScCRsrpThld set to a value other than 255	<i>EPC-based NSA Basics</i>	When NSA/SA selection based on uplink coverage takes effect, the function of FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs does not take effect.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all boards that support NSA networking support this function. For details about the boards that support NSA networking, see the hardware requirements of NSA networking based on EPC in *EPC-based NSA Basics*.
- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.3.4 Networking

The NSA networking must use the Option 3x architecture.

5.3.5 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later and support the following capabilities:
 - SA capability
 - **handoverLTE-EPC** capability
 - EN-DC capability (whether the UE whose **RAT-Type** is "eutra" supports "en-DC-r15" capability); the band to which the current NR cell belongs contained in the *supportedBandListEN-DC-r15* IE in the UE capability
- Core network requirements
 - The core network must comply with 3GPP Release 15 or later.
 - The N26 interface must be already configured between the MME and AMF.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-3](#), [Table 5-4](#), and [Table 5-5](#) describe the related parameter settings for NSA/SA selection based on user experience, with recommended values for different services and types of UEs provided in [Table 5-6](#) and [Table 5-7](#). For details about the basic parameters for mobility from NG-RAN to E-UTRAN, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* > "Data Preparation" and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* > Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode > "Operation and Maintenance (NG-RAN)" > "Data Preparation".

Table 5-3 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Networking Option Opt Sw	gNodeBParam.N etworkingOption OptSw	Select the NSA_SA_UL_SEL_OPT_SW option.
Operator Inter-RAT Policy Switch	gNBOperator.Op eratorInterRatPol icySw	Select the NSA_SA_UL_SEL_OPT_SW option.
NSA/SA Selection Opt Period	gNodeBParam.Ns aSaSelOptPeriod	The default value is recommended.
Networking Option Opt Flag	gNBQciBearer.Ne tworkingOption OptFlag	It is recommended that the parameter settings for different QCLs on the gNodeB be consistent with those for the corresponding QCLs on the eNodeB. The parameter on the eNodeB side is ServiceIrHoCfgGroup.InterRatHoStat e . For details about the recommended settings, see Table 5-6 .

Parameter Name	Parameter ID	Setting Notes
Networking Option Opt Flag	gNBRfspConfig.NetworkingOptionOptFlag	This parameter must be set if flexible user steering is enabled and different network architecture selection policies are required for UEs with different RFSP indexes. It is recommended that the policies configured for different UE types on the gNodeB be consistent with those configured for the corresponding UE types on the eNodeB. The selection policy on the eNodeB side is controlled by the NSA_SA_SEL_OPT_SW option of the SpidCfg.SpidAlgoSwitch parameter. For details about the recommended settings, see Table 5-7.
Handover Mode switch	NRInterRatHoParam.HoModeSwitch	Select the EUTRAN_HO_SWITCH option.
Inter-RAT Handover Policy Switch	NRCellOpPolicy.InterRatHoPolicySwitch	In RAN sharing with common carrier scenarios (where the gNBSharingMode.gNBMultiOpSharingMode parameter is set to SHARED_FREQ), if the inter-RAT handover policy needs to be configured on a per operator basis, select the EUTRAN_HO_SWITCH option.
Inter-RAT Service Mobility Switch	NRCellAlgoSwitch.InterRatServiceMobilitySw	To use virtual grids for NSA/SA selection based on user experience, select the NSA_SA_SEL_OPT_RSRP_PRED_SW option.
Networking Option Opt Sw	gNodeBParam.NetworkingOptionOptSw	It is recommended that the NSA_SA_SEL_INTER_GNB_CA_SW option be selected in inter-gNodeB CA scenarios.
Mobility HO Control Switch	NRInterRatHoParam.MobilityHOControlSwitch	To disable SA-to-NSA switching based on weak uplink coverage in a cell, select the NSA_SA_UL_WEAK_FRBD_SW option for the cell.

Table 5-4 NR parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
NR Move to E-UTRAN UL SINR Low Threshold	NRDUCellSrsMeas.NrToEutranSinrLowThld	Set this parameter based on the network plan. - A larger value of this parameter results in a higher probability of switching from SA to NSA based on the uplink signal quality of the NR cell but also a higher probability of ping-pong switching. - A smaller value of this parameter results in a lower probability of switching from SA to NSA based on the uplink signal quality of the NR cell and therefore a lower probability of achieving the optimal uplink experience. The data split method is MCG split after SA-to-NSA switching. The uplink fallback to LTE function may change the method to SCG split. To prevent this change, set parameters based on network conditions. It is recommended that parameter settings meet the following condition: NRDUCellSrsMeas.NrToEutranSinrLowThld – 2 dB < NRDUCellSrsMeas.NsaUlFackToLteSinrThld + NRDUCellSrsMeas.NsaUlFackToLteSinrHyst.
Service Function Switch	NRCelAlgoSwitch.ServiceFunctionSwitch	It is recommended that the VOLUME_IDENTIFY_SW option be selected.
Common UL Large Pkt Ident Thld	NRDUCellServExp.CommonULLargePktIdentThld	Set this parameter based on the network plan. It is recommended that this parameter be set to 80 kbits. - The larger the value of this parameter, the lower the probability of switching from SA to NSA. - The smaller the parameter value, the higher the probability.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN To NR SINR High Thld	NRDUCellSrsMeas.EutranToNrSinrHighThld	Set this parameter based on the network plan. - A larger value of this parameter results in a lower probability of switching from NSA to SA based on the uplink signal quality of the NR cell and therefore a lower probability of achieving the optimal uplink experience. - A smaller value of this parameter results in a higher probability of switching from NSA to SA based on the uplink signal quality of the NR cell but also a higher probability of ping-pong switching.
NSA UL Path Select SINR Time-to-Trigger	NRDUCellSrsMeas.NsaULPathSelSinrTimeToTrig	The default value is recommended.
NSA PCC Frequency Priority	NRCCellEutranNFr eq.NsaPccFreqPriority	The value of this parameter must be the same as that of the eNodeB parameter PccFreqCfg.NsaPccAnchoringPriority.
ANR Algorithm Switch	NRCCellAnr.AnrAlgorithmSwitch	It is recommended that the NSA_SA_SEL_MISS_CELL_ANR_SW option be selected when UEs need to be handed over from SA to the neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor relationships with that cell.
ANR Mode Switch	NRCCellAnr.AnrModeSwitch	Set this parameter based on the network plan when UEs need to be handed over from SA to the neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor relationships with that cell.
Networking Option Opt RSRP Threshold	NRCCellHoEutranMeaGrp.NetworkingOptionOptB1Thld	Set this parameter based on the network plan.
Uplink Traffic Volume Threshold	CellMultiCarrUniSch.ULTrafficVolumeThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-gNodeB Exp Evaluation Coefficient	NRCellCaMgmtConfig . <i>InterGnbExpEvalCoeff</i>	Set this parameter based on the network plan.
Process Switch	NRCellAlgoSwitch . <i>ProcessSwitch</i>	It is recommended that the UE_CAP_SEPARATE_QUERY_SW option be selected if the gNodeB is required to query the NR, EUTRA-NR, and EUTRA capabilities of UEs separately. If UE capability separate query is not required for specific UEs, select the UE_CAPB_SEPARATE_QUERY_SW_OFF option of the gNBUECompatSw . <i>BlacklistCtrlSwitch</i> parameter.

Table 5-5 LTE parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	EnodebAlgoExtSwitch . <i>MultiNetworkingOptionOptSw</i>	Select the LTE_FDD_NSA_SA_UL_SEL_OPT_SW option.
Handover Mode switch	CellHoParaCfg . <i>HoModeSwitch</i>	Select the NrHoSwitch option.
Handover Mode switch	CellOpHoCfg . <i>HoModeSwitch</i>	In RAN sharing with common carrier scenarios, select the HANDOVER_TO_NR_SW option.
Handover to NR Algo Switch	EnodebAlgoExtSwitch . <i>NrHandoverAlgoSwitch</i>	Select the DIRECT_NR_HO_TO_EN_DC_SW option.
SPID Algorithm Switch	SpidCfg . <i>SpidAlgoSwitch</i>	This parameter must be set if flexible user steering is enabled and different network architecture selection policies are required for UEs with different SPIDs. Select the NSA_SA_SEL_OPT_SW option for the SPID of UEs that need to support the network architecture selection function.
SA to NSA PCC Uplink PRB Usage Threshold	NsaDcMgmtConfig . <i>SaToNsaPccUlpPrbUsageThld</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
SA to NSA PCC Active UE Number Threshold	NsaDcMgmtConfig.SaToNsaPccActiveUeNumThld	Set this parameter based on the network plan.
SA to NSA PCC Uplink Interference Threshold	NsaDcMgmtConfig.SaToNsaPccUplinkInterfThld	Set this parameter based on the network plan.
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	Select the UL_TFC_NR_LTE_MOBILITY_ENH_SW option when the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is required.

Table 5-6 Recommended settings for different services

Typical Services and Their Policies	Recommended Value of gNBQciBearer.NetworkingOptionOptFlag on the gNodeB	Recommended Value of ServiceIrrHoCfgGroup.InterRatHoState on the eNodeB
Enhanced Mobile Broadband (eMBB) services, such as QCI 9 services: Data split in NSA is recommended. SA-to-NSA switching based on user experience is allowed.	PERMIT	PERMIT_HO
Ultra-reliable low-latency communication (URLLC) services: They can be carried only in NR SA networks. Therefore, when URLLC services are initiated, UEs must return to NR SA networks as soon as possible.	FORBID	MUST_HO
Voice services, such as QCI 1 services: Set the policy as required.	FORBID	NO_HO is recommended if it is expected to reduce the number of handovers for voice services. PERMIT_HO or MUST_HO is recommended if VoNR is preferred.

Typical Services and Their Policies	Recommended Value of gNBQciBearer.NetworkingOptionOptFlag on the gNodeB	Recommended Value of ServiceIrHoCfgGroup.InterRatHoState on the eNodeB
Wireless broadband (WBB) services: Mobility reduction is recommended.	FORBID	PERMIT_HO
Other types of services, such as QCI 5 services: Use the follow-up policy.	FOLLOW	PERMIT_HO

Table 5-7 Recommended settings for different types of UEs

UE Type	Recommended Value of gNBRfspConfig.NetworkingOptionOpt-Flag on the gNodeB	Recommended Value of SpidCfg.SpidAlgoSwitch on the eNodeB
Common UEs	TRUE	NSA_SA_SEL_OPT_SW selected
Special UEs	FALSE	NSA_SA_SEL_OPT_SW deselected

5.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must be already completed before activating this function. For details, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode* > "Using MML Commands" and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* > "Basic Mobility Management Procedure for UEs in Connected Mode" > "Operation and Maintenance (NG-RAN)" > "Using MML Commands".

Activation Command Examples

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//On the gNodeB side
//Enabling NSA/SA selection based on uplink coverage
```

```

MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_UL_SEL_OPT_SW-1, NsaSaSelOptPeriod=60;
ADD GNBOPERATOR: OperatorId=1, OperatorName="PRIMARY OPERATOR", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_UL_SEL_OPT_SW-1;
//Turning on the E-UTRAN handover switch
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=EUTRAN_HO_SWITCH-1;
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=1, InterRatHoPolicySwitch=EUTRAN_HO_SWITCH-1;
//Deselecting the NSA_SA_UL_WEAK_FRBD_SW option for a target cell to enable SA-to-NSA switching
based on weak uplink coverage for this cell
MOD NRINTERRATHOPARAM: NrCellId=0, MobilityHOControlSwitch=NSA_SA_UL_WEAK_FRBD_SW-0;
//Optional Setting NetworkingOptionOptFlag if network architecture selection policies need to be
configured for different QCI
MOD GNBQCIBEARER: Qci=1, NetworkingOptionOptFlag=FORBID;
MOD GNBQCIBEARER: Qci=9, NetworkingOptionOptFlag=PERMIT;
//Optional; required only when RSRP prediction based on virtual grids is to be used) Enabling RSRP
prediction for NSA/SA selection optimization
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=NSA_SA_SEL_OPT_RSRP_PRED_SW-1;
//Optional; required only when flexible user steering is enabled) Setting the network architecture selection
policy for an RFSP index
MOD GNBRSFSPCONFIG: OperatorId=1, RfspIndex=1, NetworkingOptionOptFlag=TRUE;
//Optional, required only in inter-gNodeB CA scenarios) Turning on the NSA/SA Sel for Inter-gNodeB CA
Switch
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-1;
//Optional, required only in inter-gNodeB CA scenarios) Setting the coefficient used for experience
evaluation in inter-gNodeB scenarios
MOD NRCELLCAMGMTCONFIG: NrCellId=0, InterGnbExpEvalCoeff=70;

//On the eNodeB side
//Enabling NSA/SA selection based on uplink coverage
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_UL_SEL_OPT_SW-1,
NrHandoverAlgoSwitch=DIRECT_NR_HO_TO_EN_DC_SW-1;
//Turning on the NR handover switch
MOD CELLHOPARACFG: LocalCellId=0, HoModeSwitch=NrHoSwitch-1;
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=1, HoModeSwitch=HANDOVER_TO_NR_SW-1;
//Optional Setting service-based inter-RAT handover policies if network architecture selection policies
need to be configured for different QCIs (The policies must be consistent with those configured on the
gNodeB. Otherwise, a ping-pong effect will occur.)
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=1, ServiceHoNrPolicyGroupId=0;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, ServiceHoNrPolicyGroupId=1;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=0, InterRatHoState=NO_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=1, InterRatHoState=PERMIT_HO;
//Optional; required only when flexible user steering is enabled) Setting the network architecture selection
policy for an SPID. The policy must be consistent with that configured on the gNodeB.
MOD SPIDCFG: Spid=1, SpidAlgoSwitch=NSA_SA_SEL_OPT_SW-1;
//Running the following commands for LTE cells if the function of prohibiting large-packet UEs in LTE cells
from returning to NR cells is required
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=UL_TFC_NR_LTE_MOBILITY_ENH_SW-1;
MOD CELLMULTICARRUNISCH: LocalCellId=0, ULTrafficVolumeThld=100;

```

Optimization Command Examples

```

//On the gNodeB side
//Changing the settings of NrToEutranSinrLowThld, EutranToNrSinrHighThld, and
NsaUlPathSelSinrTimeToTrig
MOD NRDUCELLSRSEAS: NrDuCellId=0, NrToEutranSinrLowThld=-32, EutranToNrSinrHighThld=160,
NsaUlPathSelSinrTimeToTrig=10;
//Setting the NSA PCC frequency priority
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLearfcn=1300, NsaPccFreqPriority=10;
//Optional Run the following commands if UEs are required to be handed over from SA to the
neighboring LTE cell with the highest anchor priority and RSRP but the gNodeB does not have neighbor
relationships with this neighboring cell
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NSA_SA_SEL_MISS_CELL_ANR_SW-1;
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_EUTRAN_ANR_CTRL_MODE_SW-0;
//Setting the RSRP threshold for event B1 related to network architecture selection
MOD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
NetworkingOptionOptB1Thld=-111;
//Optional Turning on the traffic volume identification switch if data volume needs to be considered for
this function to take effect
MOD NRCELLALGOSWITCH: NrCellId=0, ServiceFunctionSwitch=VOLUME_IDENTIFY_SW-1;

```

```

//(Optional) Changing the setting of the uplink traffic volume threshold based on the network plan
MOD NRDUCELLSERVEXP: NrDuCellId=0, CommonULLargePktIdentThld=50;
//(Optional) Selecting the UE_CAP_SEPARATE_QUERY_SW option if the gNodeB is required to query the NR,
EUTRA-NR, and EUTRA capabilities of UEs separately
MOD NRCELLALGOSWITCH: NrCellId=0, ProcessSwitch=UE_CAP_SEPARATE_QUERY_SW-1;
//(Optional) Selecting the UE_CAPB_SEPARATE_QUERY_SW_OFF option when UE capability separate query
is not needed for specific UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlSwitch=UE_CAPB_SEPARATE_QUERY_SW_OFF-1;

//On the eNodeB side
//Changing the settings of SaToNsaPccActiveUeNumThld, SaToNsaPccULIntrfThld, and
SaToNsaPccULPrbUsageThld
MOD NSADCMGMTCONFIG: LocalCellId=0, SaToNsaPccActiveUeNumThld=60, SaToNsaPccULIntrfThld=-113,
SaToNsaPccULPrbUsageThld=40;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivating NSA/SA selection based on uplink coverage for the base station

```

//On the gNodeB side
//(Optional, required only in inter-gNodeB CA scenarios) Turning off the NSA/SA Sel for Inter-gNodeB CA
Switch
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-0;
//(Optional) Disabling RSRP prediction based on virtual grids
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=NSA_SA_SEL_OPT_RSRP_PRED_SW-0;
//Disabling NSA/SA selection based on uplink coverage
MOD GNBOPERATOR: OperatorId=1, OperatorName="PRIMARY OPERATOR", Mcc="460", Mnc="00",
OperatorInterRatPolicySw=NSA_SA_UL_SEL_OPT_SW-0;
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_UL_SEL_OPT_SW-0;

//On the eNodeB side
//Disabling NSA/SA selection based on uplink coverage
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_UL_SEL_OPT_SW-0;

```

Deactivating SA-to-NSA switching based on weak uplink coverage for a cell

```

//Selecting the NSA_SA_UL_WEAK_FRBD_SW option to disable SA-to-NSA switching based on weak uplink
coverage for the desired cell
MOD NRINTERRATHOPARAM: NrCellId=0, MobilityHOCtrlSwitch=NSA_SA_UL_WEAK_FRBD_SW-1;

```

Deactivating ANR for missing neighboring LTE cells for a cell

```

//Running the following command if UEs are not expected to be handed over from SA to the neighboring
LTE cell with the highest anchor priority and RSRP and the gNodeB does not have neighbor relationships
with this neighboring cell
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NSA_SA_SEL_MISS_CELL_ANR_SW-0;

```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

[Using Signaling Tracing](#) and [Using Counters](#) can be used to verify whether SA-to-NSA switching based on weak uplink coverage has taken effect.

Using Signaling Tracing

To verify whether SA-to-NSA switching based on weak uplink coverage has been activated, perform the following steps:

- Step 1** Start Uu, NG, and X2 interface tracing for the gNodeB.
- Step 2** Start X2 interface tracing for the eNodeB.
- Step 3** Observe X2 messages on the gNodeB side. If X2 messages include an EN-DC Private Evaluation Request message that carries the *Evaluation Direction Indication* field, the resource evaluation request message for SA-to-NSA switching based on weak uplink coverage has been successfully sent.
- Step 4** Observe Uu messages on the gNodeB side. If the value of the *targetRAT-Type* IE in the MobilityFromNRCommand message sent by the gNodeB to the relevant UE is "eutra", the preparation for the handover from NG-RAN to E-UTRAN has succeeded.
- Step 5** Observe NG messages on the gNodeB side. If the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the *Successful Handover* IE from the AMF, the handover from NG-RAN to E-UTRAN has been successfully executed.
- Step 6** Observe X2 messages on the eNodeB side. If there is an SgNB Addition Request Acknowledge message, an SCG has been successfully added.

----End

To verify whether NSA/SA selection with RSRP predicted based on virtual grids has taken effect, check the number of outgoing handover attempts based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback), which equals **N.HO.LnrVirtualGrid.Model.PrepAttOut** minus **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut**. If the number is larger than zero, NSA/SA selection with RSRP predicted based on virtual grids has taken effect.

Using Counters

If any of the counters listed in [Table 5-8](#) is not 0, SA-to-NSA switching based on weak uplink coverage has taken effect.

Table 5-8 Counters related to SA-to-NSA switching based on weak uplink coverage

Counter ID	Counter Name
1911853806	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.UL
1911853805	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.UL
1911853804	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.UL

5.4.3 Network Monitoring

To evaluate the performance of NSA/SA selection based on UE measurements, monitor the counters listed in [Table 5-9](#) and [Table 5-10](#).

Table 5-9 Counters used to monitor SA-to-NSA switching based on UE measurements

Counter ID	Counter Name
1911829365	N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
1911829366	N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
1911829367	N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
1526765175	L.IRATHO.N2E.Direct2EnDc.PrepAttIn
1526765176	L.IRATHO.N2E.Direct2EnDc.PrepInSucc
1526765179	L.IRATHO.N2E.Direct2EnDc.ExecSuccIn

Table 5-10 Counters used to monitor NSA-to-SA switching based on UE measurements

Counter ID	Counter Name
1526765180	L.IRATHO.E2N.NrULCoverage.PrepAttOut
1526765177	L.IRATHO.E2N.NrULCoverage.ExecAttOut
1526765178	L.IRATHO.E2N.NrULCoverage.ExecSuccOut

To evaluate the performance of NSA/SA selection with RSRP predicted based on virtual grids, monitor the results of the following formulas:

- Number of successful handovers based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.ExecSuccOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecSuccOut**
- Number of outgoing handover executions based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.ExecAttOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecAttOut**
- Number of outgoing handover attempts based on the values predicted using LTE and NR inter-RAT virtual-grid-based prediction models (excluding EPS fallback) = **N.HO.LnrVirtualGrid.Model.PrepAttOut** – **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut**

Table 5-11 NR counters used to monitor NSA/SA selection with RSRP predicted based on virtual grids

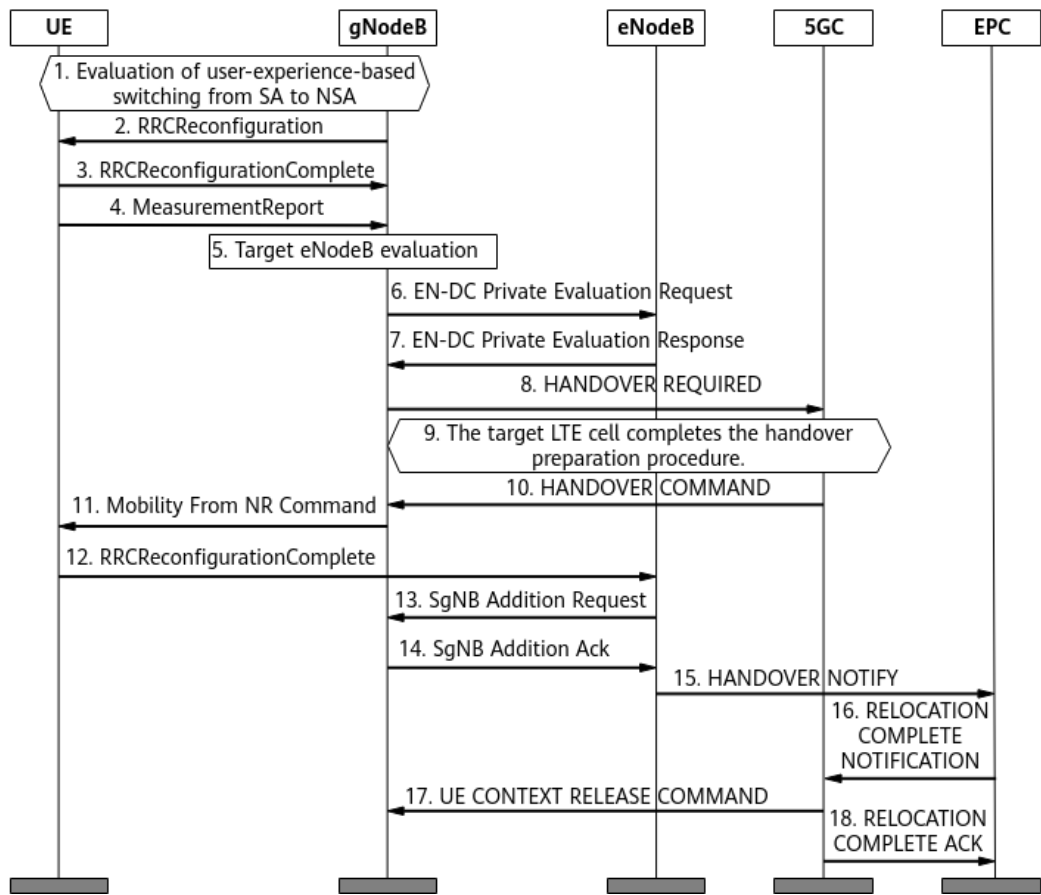
Counter ID	Counter Name
1911831568	N.HO.LnrVirtualGrid.Model.ExecSuccOut
1911831569	N.HO.LnrVirtualGrid.Model.ExecAttOut
1911831570	N.HO.LnrVirtualGrid.Model.PrepAttOut
1911831561	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecSuccOut
1911831562	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecAttOut
1911831563	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut

6 Appendix

6.1 NSA/SA Selection Based on User Experience

Figure 6-1 shows the messages related to the NSA/SA selection based on user experience.

Figure 6-1 Message flow for NSA/SA selection based on user experience



The procedures are as follows:

1. The gNodeB determines to start user-experience-based switching from SA to NSA.
2. The gNodeB delivers the measurement configurations related to inter-RAT event B1.
3. The UE responds that the measurement configuration is complete.
4. The UE sends measurement reports on inter-RAT event B1.
5. The gNodeB selects the target eNodeB.
6. The gNodeB sends an available resource evaluation assistance message to the target eNodeB over the X2 interface.
7. The target eNodeB sends the evaluation result to the gNodeB over the X2 interface.
8. The gNodeB initiates a handover request.
9. The target LTE cell completes the handover preparation procedure, which involves the following messages:
 - a. AMF to MME: RELOCATION REQUEST
 - b. MME to eNodeB: HANDOVER REQUEST
 - c. eNodeB to MME: HANDOVER REQUEST ACK
 - d. MME to AMF: RELOCATION RESPONSE
10. The 5GC notifies the gNodeB that the handover preparation is complete.
11. The gNodeB initiates handover control towards the UE.
12. The UE sends an RRC connection reconfiguration complete message.
13. The eNodeB instructs the gNodeB to add an SCG.
14. The gNodeB sends a message, indicating that the SCG has been added.
15. The eNodeB informs the EPC that the handover is complete.
16. The EPC notifies the 5GC that the data forwarding tunnel has been established.
17. The gNodeB receives a UE context release message and then releases the UE context.
18. The 5GC responds to the EPC that data forwarding is complete.

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Experience Improvement*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *Flexible User Steering in eRAN Feature Documentation*
- *Carrier Aggregation in eRAN Feature Documentation*
- *Multi-Operator Sharing in 5G RAN Feature Documentation*
- *Flexible User Steering in 5G RAN Feature Documentation*
- *Multi-Frequency Smart Selection in 5G RAN Feature Documentation*
- *Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation*
- *Licensed Assisted Access (TDD) in eRAN Feature Documentation*
- *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD) in eRAN Feature Documentation*
- *License Control Item Lists in 5G RAN Feature Documentation*
- *ANR in 5G RAN Feature Documentation*
- *Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation*
- *3900 & 5900 Series Base Station Product Documentation*
- *3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"*